Monodromy of rational singularities on surfaces in mixed characteristic

Abstract

We study the ℓ-adic monodromy of smooth proper surfaces over complete discrete valuation fields of mixed characteristic $(0, p)$, with algebraically closed residue field and $p > 5$. When the surfaces admit proper models whose special fiber has a rational double point, we show that the monodromy action of inertia is finite, depends on an étale neighborhood of the singularity and its nontrivial part coincides with a restriction of the reflection representation of a Weyl group naturally associated with the singularity. We extend this description for degenerations into general rational singularities whose deformations lie in the Artin component: the relevant Weyl group is generated by the exceptional (-2)-curves in the minimal resolution of the special fiber. Moreover we determine minimal finite extensions over which such surfaces have good reduction. The proof relies on a mixed-characteristic extension of Grothendieck–Springer simultaneous resolutions and an identification of monodromy and Springer representations via relatively perverse nearby cycles.

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1. Introduction

Let Λ be a henselian discrete valuation ring of mixed characteristic $(0, p)$, by which we mean that $K = \mathrm{Frac}(\Lambda)$ has characteristic zero and residue field k is perfect of characteristic $p > 0$. Given a smooth proper variety X/K with a proper flat model $\mathcal{X}/\Lambda$, a central problem is to relate the geometry of the special fiber $\mathcal{X}_k$ to the action of inertia I_K on $\mathrm{H}^i_{\text{ét}}(X_{\overline{K}}, \mathbb{Q}_\ell)$ for $\ell \neq p$. Here $I_K = \ker(\mathrm{Gal}_K \twoheadrightarrow \mathrm{Gal}_k)$ and its action on cohomology is referred to as the *ℓ-adic monodromy* action.

If $\mathcal{X}$ is smooth over Λ then by smooth proper base-change we have that the I_K-action is trivial, i.e. all $\mathrm{H}^i_{\text{ét}}(X_{\overline{K}}, \mathbb{Q}_\ell)$ are unramified Gal_K-representations. However, unramified cohomology does not imply that X admits a smooth proper model after extension. For special classes of varieties, trivial monodromy action is equivalent to good reduction: if X is an abelian variety, then by the Néron–Ogg–Shafarevich criterion X admits a smooth proper model over Λ if and only if $\mathrm{H}^1_{\text{ét}}(X_{\overline{K}}, \mathbb{Q}_\ell)$ is unramified ([ST68, Thm. 1]).

For general varieties, Grothendieck's local monodromy theorem asserts that I_K acts quasi-unipotently, i.e. there is an open subgroup $U \subset I_K$ so that $g \in U$ acts as $\exp(t_\ell(g)N)$ where t_ℓ is the tame Kummer character and N is the monodromy operator [SGA 7_{I}, Exposé I, Th. 1.2]. When X has strictly semistable reduction over Λ, I_K acts unipotently and Rapoport–Zink have given a geometric description of N via nearby cycles and a spectral sequence whose terms are the cohomology of intersections of components of the special fiber ([RZ82]).

The study of nearby cycles has also been fruitful in the non-semistable case when $\mathcal{X}_k$ has ordinary double point singularities: in [SGA 7_{II}] the ℓ-adic monodromy action is computed via Picard–Lefschetz formulas using vanishing cycle classes. These results were subsequently used in the study of Lefschetz pencils and Deligne's proof of the Weil conjectures ([Del74, §4.3 and §5.8]).

A more recent arithmetic application of monodromy concerns good reductions of K3 surfaces: for a K3 surface X and assuming the existence of a potential Kulikov model, Liedtke–Matsumoto show that $\mathrm{H}^2_{\text{ét}}(X_{\overline{K}}, \mathbb{Q}_\ell)$ unramified implies the existence of a projective model $\mathcal{X}/\Lambda$ with $\mathcal{X}_k$ having at worst rational double point singularities, and X has good reduction after a finite *unramified* extension of K ([LM18, Thm. 1.3]). These results motivate the direct description of ℓ-adic monodromy arising from degenerations to varieties with a prescribed set of singularities, and the arithmetic implications on good reduction of X.

In this paper we describe the monodromy action on $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ for surface degenerations with rational double points (of which ordinary double points are the simplest

case), and more generally with rational singularities whose deformations lie in their Artin component. In the spirit of [SGA 7_{II}] and [Ill94], we use nearby cycles and more recent tools from perverse sheaves and Springer theory to determine the intrinsic I_K-action on $\mathrm{H}^2_{\text{ét}}(X_{\overline{K}}, \mathbb{Q}_\ell)$ and detect good reduction criteria in this local setting.

1.1 Main results. Suppose now that Λ is complete and residue field k is algebraically closed. Let X/K be a smooth proper surface with a proper flat model $\mathcal{X}/\Lambda$ so that $\mathcal{X}_k$ has a unique isolated rational double point singularity x.

Rational double points on surfaces are characterized as normal singularities whose exceptional divisor in the minimal resolution is a tree of rational curves of self-intersection -2 (see Definition 2.1 for details). Such singularities are classically also known as Kleinian, Du Val or simple singularities, see [Dur79] for a thorough survey of their literature.

There is an important link between rational double points and ADE Dynkin diagrams: if $E_1, \cdots, E_r$ denote the exceptional curves in the minimal resolution $\widetilde{\mathcal{X}}_k$, then under the intersection pairing on $\mathrm{Pic}(\widetilde{\mathcal{X}}_k)$, $\Phi = \bigoplus_i \mathbb{Z}[E_i]$ forms an irreducible root lattice of Dynkin type A, D or E, with simple roots $[E_i]$. Under suitable restrictions on p, the Dynkin diagram of Φ (equivalently the dual graph of x) determines the formal isomorphism class of the singularity (see Section 2.1 for more; $p > 5$ here suffices). Our first theorem is as follows:

Theorem 1.1 (Theorem 5.16)**.** *Let $p > 5$ and $\mathcal{X}/\Lambda$ a proper flat surface degeneration with $X = \mathcal{X}_K$ a smooth proper surface and $\mathcal{X}_k$ having a unique rational double point x. Let Φ be its associated root lattice, $V_\ell = \Phi^\vee \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$ and W the Weyl group of Φ.*

(i) *The monodromy I_K-action on $\mathrm{H}^i(X_{\overline{K}}, \mathbb{Q}_\ell)$ is trivial for $i \neq 2$. For $i = 2$ let L be the fixed field of the kernel of the I_K-action, then we have an I_K-equivariant splitting*

$$\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^2(\mathcal{X}_k, \mathbb{Q}_\ell) \oplus V_\ell(-1) \tag{1.1}$$

The monodromy I_K-action is trivial on the first summand, and on the second it factors through $\mathrm{Gal}(L/K)$, where it is identified with the restriction of the natural W-action on V_ℓ along an embedding $\mathrm{Gal}(L/K) \hookrightarrow W$.

(ii) *L is the minimal Galois extension over which X_K has good reduction in the category of algebraic spaces. Moreover, the isomorphism class of I_K-representation V_ℓ and the extension L/K depend only on a formal affine neighborhood of $(\mathcal{X}, x)$.*

Remark 1.2. As a consequence, if X has a rational double point reduction, then its ℓ-adic monodromy is finite (equivalently $N = 0$ for the monodromy operator). By (ii), X has potential good reduction, but the smooth algebraic space model need not be a scheme, see Remark 5.20.

We can extend Theorem 1.1 to degenerations where $\mathcal{X}_k$ has an isolated rational surface singularity, under suitable restrictions on the model $\mathcal{X}/\Lambda$. Rational singularities are characterized by the vanishing of higher pushforwards along the minimal resolution: if

$\pi : \widetilde{\mathcal{X}}_k \to \mathcal{X}_k$ is a minimal resolution then $\mathrm{R}\pi_*\mathcal{O}_{\widetilde{\mathcal{X}}_k} \simeq \mathcal{O}_{\mathcal{X}_k}$. Rational double points form a subclass of rational singularities – in general, the exceptional divisor over a rational singularity is a tree of rational curves of self-intersection ≤ -2, with equality holding when the singularity is a rational double point.

The restriction on $\mathcal{X}$ is that the induced deformation of the singularity of $\mathcal{X}_k$ lies within a special component of the deformation space called the *Artin component*. Such deformations are simultaneously resolvable in mixed characteristic by the methods of [Art74] and we may apply the arguments of Theorem 1.1 to obtain the following.

Theorem 1.3 (Theorem 6.3)**.** *Let $p > 5$ and $\mathcal{X}/\Lambda$ be a proper flat surface degeneration with $X = \mathcal{X}_K$ a smooth proper surface and $\mathcal{X}_k$ having a unique rational singularity so that $\mathcal{X}$ induces a deformation within its Artin component. Let Φ be the exceptional curve lattice inside $\mathrm{Pic}(\widetilde{X}_k)$ and let $\Psi \subset \Phi$ be the root sub-lattice generated by the exceptional curves of self-intersection -2, $W = W(\Psi)$ the associated Weyl group. Set $V_\ell = \Phi^\vee \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$ and $N_\ell = \Psi^\vee \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$, and view $N_\ell \subset V_\ell$ via the intersection pairing. Then the monodromy I_K-action on $\mathrm{H}^i(X_{\overline{K}}, \mathbb{Q}_\ell)$ is trivial for $i \neq 2$ and for $i = 2$ we have an I_K-equivariant splitting*

$$\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^2(\mathcal{X}_k, \mathbb{Q}_\ell) \oplus V_\ell(-1)$$

The I_K-action is trivial on the first summand, and on the second summand it acts on $V_\ell = N_\ell \oplus N_\ell^\perp$ via restriction along a homomorphism $\rho : I_K \to W$, where W acts on N_ℓ via reflections and on $N_\ell^\perp$ it acts trivially. Moreover if L is the fixed field of $\ker \rho$ then L/K is the minimal finite Galois extension over which X has good reduction.

The Artin component condition is automatic for all rational double points, so Theorem 1.3 is a natural extension of Theorem 1.1. We remark however that for deformations of rational singularities outside the Artin component we do not know whether the ℓ-adic cohomological monodromy is finite; for projective surface degenerations over $\mathbb{C}$, the answer is positive using tools from Hodge theory (see [KLS22], Cor. 1).

Our results naturally extend to $\mathcal{X}_k$ having finitely many rational singularities (cf. Remark 5.17). One observation is that if X admits a semistable model over Λ in addition to a rational double point model, then it has good reduction over K since a finite unipotent monodromy action is trivial.

Another observation is that if $p \nmid |W|$, then the good reduction extension L/K is tame and thus $\mathrm{Gal}(L/K)$ is generated by a single element T; we can describe the monodromy I_K-action via this *monodromy operator* T. In this setting we recover a result of Liedtke–Matsumoto on tame extensions of K3 surfaces and good reduction ([LM18, Cor. 1.5]): if X is a K3 surface with a rational double point model and $p \geq 23$, then since the maximal number of exceptional curves is bounded by 21 one gets that $p \nmid |W|$ and X has good reduction after a tame extension.

The simplest example of Theorem 1.1 is the case of ordinary double point reduction, which has an elementary explanation.

Example 1.4 (Ordinary double points)**.** Let $\pi \in \Lambda$ be a uniformizer and consider the family of quadrics

$$X_n = \{z^2 + xy = \pi^n w^2\} \subset \mathbb{P}^3_\Lambda, \quad n \geq 1$$

The special fiber $(X_n)_k = \{z^2 + xy = 0\}$ has an ordinary double point and the generic fiber $(X_n)_K$ is K-isomorphic to $(X_2)_K$ or $(X_1)_K$ depending on the parity of n.

Set $t = \pi^{n/2} \in \overline{K}$, $L = K(t)$, then $(X_n)_L \simeq \mathbb{P}^1_L \times \mathbb{P}^1_L$. Let $a, b \in \mathrm{H}^2((X_n)_{\overline{K}}, \mathbb{Q}_\ell(1))$ be the classes of the two lines $\{x = 0, z = \pm tw\}$ in $(X_n)_L$. Inertia fixes $a + b$ and acts on $a - b$ via χ_n, where $\chi_n(g) = g(t)/t$. Thus as an I_K-representation we have

$$\mathrm{H}^2((X_n)_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathbb{Q}_\ell(-1) \oplus (\mathbb{Q}_\ell(-1) \otimes \chi_n)$$

Set $\delta = a - b$. Since $(\delta, \delta) = -2$ we can describe the I_K-action as

$$g(v) = v + \frac{1 - \chi_n(g)}{2}(v, \delta)\delta, \quad g \in I_K$$

This recovers the Picard–Lefschetz formulas in [SGA 7_{II}, Exposé XV, §3.2.1-3.2.3].

If n is even then $t \in K$ and the monodromy action on $\mathrm{H}^2((X_n)_{\overline{K}}, \mathbb{Q}_\ell)$ is trivial; in fact $(X_n)_K \simeq \mathbb{P}^1_K \times \mathbb{P}^1_K$ is already a split quadric and has good reduction over K with associated smooth proper model $\mathbb{P}^1_\Lambda \times \mathbb{P}^1_\Lambda$.

If n is odd, the minimal field where $(X_n)_K$ attains good reduction is $L = K(\sqrt{\pi})$. The character χ_n is tamely ramified when $p > 2$ and wildly ramified when $p = 2$.

1.2 Proof outline. We give a sketch of the main steps in proving Theorem 1.1. The model $\mathcal{X}/\Lambda$ induces a formal deformation $\widehat{\mathcal{O}}_{\mathcal{X},x}$ of the singularity x, and a nearby cycles argument shows that the monodromy action on $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ can be computed from a formal neighborhood of the singularity.

When the characteristic p is 'good' (cf. Section 2.1) there is a connection between rational double points x and simple simply-laced Lie algebras $\mathfrak{g}$ via identifying the dual graph of the minimal resolution of x with the Dynkin diagram of $\mathfrak{g}$. Work of many people ([Bri71], [Tju70], [Slo80a]) has expanded on this relationship and realized rational double points as the generic singularities in slices transversal to the nilpotent orbit inside the nilpotent cone $\mathcal{N} \subset \mathfrak{g}$; such a realization uniquely pins down the singularity class in 'good' characteristic.

By results of Springer and Borho–MacPherson ([Spr76], [BM83]), the Grothendieck–Springer resolution $\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ may be viewed as a simultaneous resolution of the fibers of the adjoint quotient $\mathfrak{g} \to \mathfrak{h}/\!\!/W$ after a Weyl base-change $\mathfrak{h} \to \mathfrak{h}/\!\!/W$. Restricting to appropriate slices, one can simultaneously resolve rational double points and their deformations ([Slo80b]).

We develop a mixed-characteristic incarnation of Grothendieck–Springer theory and obtain via an appropriate pullback a simultaneous resolution for our deformation $\widehat{\mathcal{O}}_{\mathcal{X},x}$; we thus get a smooth proper algebraic space $\widetilde{\mathcal{X}}$ with generic fiber X_L where L/K is the extension defined by the pullback of the Weyl cover.

The formalism of nearby cycles along with the relative perversity of the universally locally acyclic sheaf $\mathrm{R}\pi_*\mathbb{Q}_\ell[\dim \mathfrak{g}]$ (see [HS23]) allows us to identify the monodromy action on $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ with a suitable restriction of the Weyl group action on the cohomology of

the fibers of $\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$, also known as a Springer representation. This gives the required description of monodromy in Theorem 1.1.

For the general rational singularities considered in Theorem 1.3, we can partially simultaneously resolve them after a finite extension and then descend to another model $\mathcal{X}'$ over Λ whose special fiber has only rational double point singularities. We can therefore reduce the monodromy calculation to the case of Theorem 1.1.

We remark that this paper is a natural extension of the ideas in [Slo80b], which hold over fields of characteristic zero or $p \gg 0$, to the relative setting over a mixed characteristic base. Parts of this work also relate to Artin's simultaneous resolution functors ([Art74]), which were later used to extract Weyl covers as Galois covers and study the integral cohomology of partial flag varieties ([SB21]) even in bad characteristic. The simultaneous resolution of general rational singularities over $\mathbb{C}$ has also been considered by Wahl in [Wah79]. The key point for us is that good characteristic allows us to give a concrete description of Galois monodromy actions as Springer actions of the Weyl group.

Organization of the paper. Section 2 contains some background on rational singularities, simultaneous resolutions and deformations of both rational double points and their resolutions. Section 3 contains the Lie-theoretic prerequisites for the proof of Theorems 1.1 and 1.3, with a focus on the Grothendieck–Springer resolution in mixed characteristic. Section 4 uses Grothendieck–Springer theory to give a Lie-theoretic description of deformations and simultaneous resolutions of rational double points in mixed characteristic. Section 5 contains the proof of Theorem 1.1 along with technical ingredients such as nearby cycles and relative perverse sheaves. Finally Section 6 extends our results to suitable rational singularities and proves Theorem 1.3.

Notations. Throughout the paper we will assume the following conventions.

- Λ denotes a complete discrete valuation ring of mixed characteristic $(0, p)$, i.e. the fraction field $K = \mathrm{Frac}(\Lambda)$ has characteristic 0 and the residue field k has characteristic p. Unless stated otherwise we assume $\overline{k} = k$. Typical examples include $\Lambda = W(k)$ or a ramified extension $W(k)[T]/(T^n - p)$.
- By a *surface degeneration* $\mathcal{X}/\Lambda$ we mean a flat proper scheme $\mathcal{X} \to \mathrm{Spec}(\Lambda)$ with generic fiber $X = \mathcal{X}_K$ a smooth proper surface and special fiber $\mathcal{X}_k$ having a *unique* isolated singularity, or finitely many isolated singularities of a prescribed type.
- $G_K = \mathrm{Gal}_K := \mathrm{Gal}(\overline{K}/K)$ is the absolute Galois group of K. In general we have an exact sequence
$$1 \to I_K \to G_K \to G_k \to 1$$
which defines inertia I_K; when k is algebraically closed, $I_K = G_K$.
- Unless stated otherwise, all cohomologies $\mathrm{H}^i(-, \mathbb{Q}_\ell)$ are computed with respect to the small étale site.

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2. Deformations and simultaneous resolutions of rational singularities

We collect here some preparatory results on rational double points (Section 2.1), simultaneous resolutions (Section 2.2) and the deformation theory of isolated rational singularities (Sections 2.3-2.5).

2.1 Rational double points.

Throughout, we let k be an algebraically closed field of characteristic p.

Definition 2.1. Suppose X/k is an algebraic surface with isolated singularity $x \in X(k)$.

(i) x is a *rational* singularity if for its minimal resolution[1] $f : \widetilde{X} \to X$ we have

$$\mathrm{R}f_*\mathcal{O}_{\widetilde{X}} \simeq \mathcal{O}_X$$

(ii) x is a *rational double point* (henceforth an RDP) if it is normal and its minimal resolution $f : \widetilde{X} \to X$ has as reduced exceptional divisor a finite union of smooth rational curves with self intersection -2.

Rational singularities are normal since $f_*\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$, and the definitions do not depend on f. Rational double points in particular are equivalently defined as rational Gorenstein surface singularities.

A crucial fact about RDPs is that in appropriate characteristics (see Definition 2.3 below) they are *taut*, i.e. their completed local ring is determined up to isomorphism by the dual graph of their minimal resolution. This is classical over $\mathbb{C}$, and has been extended to positive characteristic by [Art77]:

Theorem 2.2 ([Art66], [Art77])**.** *Let X be an algebraic surface over k.*

(i) *If E is a connected curve on X with smooth rational components E_i so that X is smooth along E, $E_i^2 = -2$ and their intersection matrix is negative-definite, then the only possible dual graphs for E are the Dynkin diagrams A_n, D_n $(n \geq 4)$, E_6, E_7 and E_8.*

(ii) *If $x \in X(k)$ is an RDP and* $\mathrm{char}(k)$ *is good (see* Definition 2.3*) then up to analytic isomorphism $\widehat{\mathcal{O}}_{X,x} \simeq k[\![x,y,z]\!]/(f(x,y,z))$, where $f(x,y,z)$ and the dual graph of the minimal resolution of x are given by the table below:*

[1] Such a resolution always exists for surfaces in characteristic p, see e.g. [Lip78].

$f(x,y,z)$	Dual graph
$xz+y^{n+1}$	A_n
$z^2+x^2y+y^{n-1}$	D_n
$z^2+x^3+y^4$	E_6
$z^2+x^3+xy^3$	E_7
$z^2+x^3+y^5$	E_8

A polynomial $f(x,y,z)$ defining an RDP is called a 'normal form'. We note that each ADE Dynkin diagram determines a simple simply-laced root lattice Φ, and we say an RDP has 'Dynkin type Φ' when its dual graph is the Dynkin diagram of Φ.

Since the notion of 'good characteristic' will be ubiquitous in this paper, we record the definition here.

Definition 2.3. Given a semisimple simply-laced root lattice Φ of a Lie algebra of *ADE* type over a characteristic p field, we say p is:

(i) *good* with respect to Φ if $p \neq 2$ when Φ has any D_n components, $p \neq 2,3$ when Φ has any E_6, E_7 components and $p \neq 2,3,5$ when Φ has any E_8 components (note any p is good for A_n components).

(ii) *very good* with respect to Φ if it is good and for any A_n component we have $p \nmid n+1$.

(iii) *sufficiently good* with respect to Φ if $p \nmid |W(\Phi)|$ i.e. it does not divide the order of the associated Weyl group of Φ.

For semisimple root lattices, sufficiently good $\Rightarrow$ very good $\Rightarrow$ good. Via Theorem 2.2, we say p is good (or very good, or sufficiently good) for an RDP singularity if it is so for the singularity's Dynkin type. However, Theorem 2.2 (ii) is no longer true when p is *not* good, because the singularities need not be taut and different types of RDPs can have the same dual graph; see ([Art77]) for details.

2.2 Simultaneous resolutions. We next describe a particularly rare notion of resolving singularities of schemes in families. In order to be consistent with [Art74], we enlarge the category of schemes to include algebraic spaces.

Definition 2.4. Let $f : \mathcal{X} \longrightarrow S$ be a finite-type morphism of separated algebraic spaces. A *simultaneous resolution over* S is a proper S-morphism $\pi : \widetilde{\mathcal{X}} \to \mathcal{X}$ so that $f \circ \pi$ is smooth and for every geometric point $\overline{s} \to S$, the induced map $\pi_s : \widetilde{\mathcal{X}}_{\overline{s}} \to \mathcal{X}_{\overline{s}}$ is a minimal resolution.

Given a finite surjection $\psi : T \to S$, the simultaneous resolution of $f : \mathcal{X} \to S$ "after basechange along ψ" is a simultaneous resolution $\pi : \widetilde{\mathcal{X}} \to \mathcal{X} \times_S T$ over T; the induced map $\widetilde{\mathcal{X}} \to S$ will not be smooth in general when ψ is ramified. When $S = \mathrm{Spec}(k)$, we recover the usual notion of a minimal resolution $\pi : \widetilde{\mathcal{X}} \to \mathcal{X}$ of algebraic spaces over k ([Stacks, Tag 0BHV]).

Example 2.5. Simultaneous resolutions rarely exist. Assume $p \neq 3$ and suppose

$$X = \{x^3 + y^3 + z^3 = tw^3\} \subset \mathbb{P}^3_k \times \mathbb{A}^1_t$$

with f being the projection to $\mathbb{A}^1_t$. The fibers over $t \neq 0$ are smooth cubic surfaces with $\chi(\mathcal{O}) = 1$ while X_0 is the projective cone over elliptic curve $\{x^3 + y^3 + z^3 = 0\}$. The blowup of the isolated vertex singularity yields a $\mathbb{P}^1$-bundle $\widetilde{X_0}$ over the elliptic curve, whence $\chi(\mathcal{O}_{\widetilde{X_0}}) = 0$. Any other resolution of X_0 also has $\chi(\mathcal{O}) = 0$ as it is a birational invariant of smooth proper surfaces (see e.g. [Har77, Prop. V.3.4]).

Hence, a simultaneous resolution of $X/\mathbb{A}^1_t$ cannot exist after finite base-change; note after a degree 3 basechange the smooth part of the family becomes isotrivial, and yet the obstruction to simultaneously resolving the family remains.

Simultaneous resolutions have been shown to exist for surface families with RDPs, both in the complex-analytic and the algebraic setting:

Theorem 2.6 ([Bri66],[Tju70], [Art74])**.** *Let $S = \mathrm{Spec}(R)$ for R an excellent normal strictly henselian local ring, and $f : X \to S$ a finitely presented flat morphism whose geometric fibers are normal surfaces with at most rational double points. Assume the relative singular locus of f is finite over S. Then f admits a simultaneous resolution after a finite basechange $\psi : T \to S$, where T is the spectrum of a normal henselian local ring and ψ can be chosen to be étale over the locus of S where f is smooth. In other words we have a commutative diagram*

$$\begin{array}{ccccc} \widetilde{X} & \xrightarrow{\pi_1} & X \times_S T & \xrightarrow{\pi_2} & X \\ & \searrow^{\widetilde{f}} & \downarrow & \square & \downarrow f \\ & & T & \xrightarrow{\psi} & S \end{array}$$

where $\widetilde{f}$ is smooth, π_1 is proper birational and π_2 is finite.

We note that the strictly henselian formulation of Theorem 2.6 gives a local statement for families over a general excellent noetherian normal base scheme S: if $f\colon X \to S$ is a family as in Theorem 2.6, then fixing a geometric point $s \to S$ and taking $S' = \mathrm{Spec}\left(\mathcal{O}^{\mathrm{sh}}_{S,s}\right), X' = X \times_S S'$, we may apply Theorem 2.6 to the induced family $X' \to S'$ and get a simultaneous resolution after a finite basechange on S'; this is the setting treated in [Art74].

Definition 2.7. In the notation of Theorem 2.6, let $\Sigma := \mathrm{Sing}(X/S)$ be the relative singular locus of f. The *discriminant locus of f* is $\Delta := f(\Sigma)_{\mathrm{red}}$, a closed subscheme of S equipped with the reduced induced subscheme structure.

Lemma 2.8. *Suppose R is a regular local ring and $f : X \to S = \mathrm{Spec}(R)$ is a flat finitely presented map with geometrically normal surfaces as fibers, so that the generic fiber is smooth, the relative singular locus Σ of f is finite over S and, étale-locally near Σ, X is a hypersurface in a smooth scheme over S of relative dimension 3. Then the discriminant locus Δ is an effective Cartier divisor.*

Proof. We may assume that locally $X = V(f)$ inside $Y = \mathrm{Spec}(R[x,y,z]^{\mathrm{sh}}_{\mathfrak{q}})$ where $\mathfrak{q} = (\mathfrak{m}_R, x, y, z)$, so that $\Sigma = \{f = f_x = f_y = f_z = 0\}$. Let η be the generic point of an irreducible component of Σ with $\mathfrak{p} = f(\eta)$, which is not the zero ideal as the generic fiber is smooth. Note η is a closed point of fiber $Y_{\mathfrak{p}}$ as Σ is finite over S, and we have $\dim \mathcal{O}_{Y,\eta} = \mathrm{ht}(\mathfrak{p}) + 3 \leq 4$ by Krull's height theorem, so $\mathrm{ht}(\mathfrak{p}) = 1$ and Δ is pure of codimension 1, hence a Cartier divisor as R is regular. □

Example 2.9. Let $S = \mathrm{Spec}(W(k))$ have residue characteristic $p > 2$ and consider maps

$$f_1 : X_1 = V(xy - z^2 + p^2) \to S, \quad f_2 : X_2 = V(xy - z^2 + p) \to S$$

Both are flat families with an A_1-singularity $\{xy - z^2 = 0\}$ as their special fiber. Let

$$Y^- = \mathrm{Bl}_{(x,z-p)}(X_1), \quad Y^+ = \mathrm{Bl}_{(x,z+p)}(X_1)$$

then $Y^{\pm}$ are distinct simultaneous resolutions of X_1: they are smooth S-schemes, over the generic fiber they are isomorphic to X_1 and over the special fiber they are both the minimal resolution of $(X_1)_k$. $Y^-, Y+$ are related by a flop $Y^+ \dashrightarrow Y^-$, obtained by blowing up the exceptional $\mathbb{P}^1_k$ of Y^+ to an exceptional divisor $\mathbb{P}^1_k \times \mathbb{P}^1_k$ and then contracting the other ruling (this is the *arithmetic* Atiyah flop).

Family f_2 does not admit a simultaneous resolution over S. If $p > 2$ then pullback along tamely ramified cover $T = \mathrm{Spec}(W(k)[p^{1/2}]) \to S$ yields a family over T of the same type as f_1, which now has two simultaneous resolutions.

When $p = 2$ we can take the wild extension $W(k)[q]/(q^2 - 2)$ to simultaneously resolve f_2. $Y^{\pm}$ are still related by a crepant flop, but it is not the Atiyah flop; blowing up the exceptional curves now yields Hirzebruch surface $\mathbb{F}_2$ instead of $\mathbb{P}^1_k \times \mathbb{P}^1_k$.

Remark 2.10. In general, one needs to consider simultaneous resolutions in the category of algebraic spaces due to global obstructions. Artin discusses an example of a K3 surface degenerating to a nodal quartic in [Art74, §1]: étale-locally at the node, we obtain two scheme-theoretic small resolutions after a base-change of degree at most 2 as in Example 2.9, but the global simultaneous resolutions are algebraic spaces that are *not* schemes (cf. Remark 5.20 too).

2.3 Formal deformations of singularities. In this section we describe versal and miniversal deformations of isolated 2-dimensional hypersurface singularities whose completed local rings are

$$R = \frac{k[\![x,y,z]\!]}{f(x,y,z)}, \quad k = \overline{k} \tag{2.1}$$

with the goal of characterizing miniversal deformations of rational double points in mixed characteristic.

Definition 2.11. Fix a complete discrete valuation ring Λ with residue field k[2] and let $\mathrm{Art}_{\Lambda,k} \subset \mathrm{Comp}_{\Lambda,k}$ denote respectively the categories of Artinian local Λ-algebras and complete noetherian local Λ-algebras of residue field k.

[2]In practice Λ will be $W(k)$ or a ramified extension of it.

(i) For $A \in \mathrm{Comp}_{\Lambda,k}$, a *formal local deformation of* R over A is a complete flat local noetherian A-algebra B with an identification $B/\mathfrak{m}_A B \simeq R$. The set of isomorphism classes of such B is denoted $\mathrm{Def}_R(A)$.

(ii) For $A \in \mathrm{Comp}_{\Lambda,k}$, $B \in \mathrm{Def}_R(A)$, an *algebraic realization* of B is a flat morphism $f : X \to \mathrm{Spec}(A)$ of finite type such that for some $x \in X_k$ we have an identification of A-algebras $\widehat{\mathcal{O}}_{X,x} \simeq B$.

When $A \in \mathrm{Art}_{\Lambda,k}$, we recover the usual notion of infinitesimal deformations of R, while for general A we get a compatible system of A_n-algebras B_n so that B_n is an infinitesimal deformation of R over A_n and $B = \varprojlim B_n$; here $A_n = A/\mathfrak{m}_A^{n+1}$ and $B_n = B/\mathfrak{m}_A^{n+1}B$.

Lemma 2.12. *Every formal local deformation B of R over $A \in Comp_{\Lambda,k}$ has the form*

$$B \simeq \frac{A[\![x,y,z]\!]}{F(x,y,z)}, \quad F \equiv f \mod \mathfrak{m}_A$$

Proof. Lifting coordinates x, y, z yields a surjection $A[\![x,y,z]\!] \twoheadrightarrow B$, with kernel I. Then since B is a flat A-algebra we have an exact sequence

$$0 \to I/\mathfrak{m}_A I \to k[\![x,y,z]\!] \to R \to 0$$

after tensoring by $\otimes_A k$. Let $F \in I$ so that $\overline{F} \in I/\mathfrak{m}_A I$ is f, then by Nakayama's lemma we have $I = (F)$. □

Definition 2.13. Put $D = k[\varepsilon]/\varepsilon^2$. The *tangent space of deformation functor* Def_R is

$$\mathrm{Def}_R(D) = \left\{ (B,i) \;\middle|\; \begin{array}{l} B \text{ flat complete Noetherian local } D\text{-algebra,} \\ i : B/\varepsilon B \xrightarrow{\sim} R \end{array} \right\} / \sim$$

and it parametrizes 1st-order deformations of R. Any $B \in \mathrm{Def}_R(D)$ has the form

$$B = B(g) \simeq \frac{D[\![x,y,z]\!]}{f + \varepsilon g}, \;\; g \in k[\![x,y,z]\!]$$

Thus an equivalent definition of the tangent space of Def_R is

$$\mathrm{Def}_R(D) \simeq T^1(R) \coloneqq k[\![x,y,z]\!]/(f, f_x, f_y, f_z), \;\; (B(g), i) \mapsto g$$

The finite dimensional k-algebra is called the *Tjurina algebra* of singularity R.

Remark 2.14. The Tjurina algebra should be distinguished from the *Milnor algebra* $k[\![x,y,z]\!]/(f_x, f_y, f_z)$ associated to the presentation of the singularity R. When $f(x,y,z)$ is weighted homogeneous of degree prime to p, one has that $f \in (f_x, f_y, f_z)$ and so the two algebras coincide, but they differ e.g. for A_{p-1} singularities.

Now let $B \in \mathrm{Def}_R(A)$ be a formal local deformation of R over A and set

$$T_\Lambda(A) = \mathrm{Hom}_k\left(\frac{\mathfrak{m}_A}{\mathfrak{m}_A^2 + \mathfrak{m}_\Lambda A}, k\right) \simeq \mathrm{Hom}_\Lambda^{\mathrm{loc}}(A, D)$$

to be the (Zariski) tangent space of the 'mod $\mathfrak{m}_\Lambda$' fiber $\mathrm{Spec}(A/\mathfrak{m}_\Lambda A)$ of $\mathrm{Spec}(A)$ (cf. [TV13, §B.1]).

Definition 2.15 ([TV13], §6)**.** For a formal local deformation $B \in \mathrm{Def}_R(A)$ of R, its *Kodaira–Spencer* map is

$$\begin{aligned} \mathrm{KS}_B : T_\Lambda(A) &\to \mathrm{Def}_R(D), \\ (v : A \to D) &\longmapsto [B \widehat{\otimes}_A D] \end{aligned}$$

where D is viewed as an A-algebra through vector v.

By Lemma 2.12, we can write $B = A[\![x, y, z]\!]/(F)$ and pick $v \in T_\Lambda A$ sending $v(F) = f + \varepsilon g_v$. Then $\mathrm{KS}_B(v) = [B(g_v)]$ in the notation of Definition 2.13. Intuitively, the Kodaira–Spencer map records infinitesimal changes of the deforming singularity along a tangent vector on the parameter space A.

Definition 2.16 ([TV13], §6)**.** Given $B \in \mathrm{Def}_R(A)$ a formal local deformation of R, and a homomorphism $\phi : A \to C$, set $B_\phi = B \widehat{\otimes}_A C$.

(i) We say B is a *versal* deformation if the following lifting property holds: given a surjection $\widetilde{C} \twoheadrightarrow C$ in $\mathrm{Art}_{\Lambda,k}$ and $\widetilde{B} \in \mathrm{Def}_R(\widetilde{C})$ so that $B_\phi \simeq \widetilde{B} \widehat{\otimes}_{\widetilde{C}} C$, there exists a homomorphism $\psi : A \to \widetilde{C}$ lifting ϕ and an isomorphism $B_\psi \simeq \widetilde{B}$ compatible with $B_\phi \simeq \widetilde{B} \widehat{\otimes}_{\widetilde{C}} C$.

(ii) We say B is a *miniversal* deformation if it is versal and the Kodaira–Spencer map KS_B is an isomorphism of k-vector spaces.

Note that in (i) we do not require lift ψ to be unique, as that would make B a universal deformation. Note also that one can take the surjections in (i) to be in $\mathrm{Comp}_{\Lambda,k}$ by [TV13, Prop. 6.21], so any formal local deformation over a complete base C is the pullback of versal deformation B along a homomorphism $A \to C$.

Proposition 2.17 ([TV13], Prop. 6.36)**.** *Suppose $A = \Lambda[\![t_1, \cdots, t_r]\!]$ is a power series ring over Λ. If $B \in \mathrm{Def}_R(A)$ is a formal local deformation of R over A and its Kodaira–Spencer map KS_B is an isomorphism, then B is a miniversal deformation.*

Proposition 2.18 ([TV13], Prop. 6.44)**.** *Let $r = \dim_k T^1(R)$ and $A = \Lambda[\![t_1, \cdots, t_r]\!]$. Choose $g_1, \cdots, g_r \in k[\![x, y, z]\!]$ whose classes in $T^1(R)$ form a k-basis, and choose lifts $\widetilde{f}, \widetilde{g}_i \in \Lambda[\![x, y, z]\!]$ of f and g_i. Then*

$$B = \frac{A[\![x, y, z]\!]}{F(x, y, z)}, \quad F = \widetilde{f} + \sum_{i=1}^{r} t_i \widetilde{g}_i$$

is a miniversal formal local deformation of R over base A.

Proof. B is a flat A-algebra by the local flatness criterion. We can identify $T_\Lambda A \simeq k^r$ with generators v_i dual to (the classes of) t_i in the sense that $v_i : A \to D = k \oplus k\varepsilon$ maps $t_j \mapsto \delta_{ij}\varepsilon$. Then, using the notation of Definition 2.15 we have

$$B_{v_i} \simeq \frac{D[\![x,y,z]\!]}{f + \varepsilon g_i}$$

so that $\mathrm{KS}_B(v_i) = [B(g_i)] = [g_i] \in T^1(R)$ upon identifying $\mathrm{Def}_R(D) \simeq T^1(R)$. Since $\{[g_i]\}$ form a basis of $T^1(R)$, we have that KS_B is an isomorphism, whence B is miniversal by Proposition 2.17. □

Example 2.19. Let $f(x,y,z) = xy - z^n$ so that R is the completed local ring of an A_{n-1}-singularity. We have

$$T^1(R) \simeq \begin{cases} k[\![z]\!]/z^{n-1}, & p \nmid n \\ k[\![z]\!]/z^n, & p \mid n \end{cases}$$

so if p is very good for A_{n-1} we have $r = n-1$ parameters and choosing $g_i = z^{n-i-1}$ yields miniversal deformation

$$B = \frac{\Lambda[\![x,y,z,t_1,\cdots,t_{n-1}]\!]}{xy - z^n + t_1 z^{n-2} + \cdots + t_{n-2}z + t_{n-1}}$$

over base $\Lambda[\![t_1,\cdots,t_{n-1}]\!]$, while if $p|n$ a miniversal deformation is

$$B = \frac{\Lambda[\![x,y,z,t_1,\cdots,t_n]\!]}{xy - z^n + t_1 z^{n-1} + t_2 z^{n-2} + \cdots + t_{n-1}z + t_n}$$

Let X/Λ be a surface degeneration so that the special fiber X_k has a unique RDP $x \in X(k)$ of type

$$\widehat{\mathcal{O}}_{X_k,x} \simeq R = \frac{k[\![x,y,z]\!]}{f(x,y,z)}$$

as in (2.1). We can view $\widehat{\mathcal{O}}_{X,x}$ as a miniversal deformation pullback as follows.

Corollary 2.20. *Let $g_1,\cdots,g_r$ be polynomial representatives of a k-basis of $T^1(R)$ and choose polynomial lifts $\widetilde{f},\widetilde{g_i} \in \Lambda[x,y,z]$ of f, g_i. Then there exist $w_i \in \mathfrak{m}_\Lambda$ so that we have an isomorphism of Λ-algebras*

$$\widehat{\mathcal{O}}_{X,x} \simeq \frac{\Lambda[\![x,y,z]\!]}{\widetilde{f} + \sum_{i=1}^r w_i \widetilde{g_i}} \tag{2.2}$$

Proof. Note that $\widehat{\mathcal{O}}_{X,x}$ is a flat Λ-algebra and $\widehat{\mathcal{O}}_{X,x} \otimes_\Lambda k \simeq R$ so it is a formal local deformation of singularity R over base Λ. Let $A = \Lambda[\![t_1,\cdots,t_r]\!]$ with $B, F(x,y,z)$ as in Proposition 2.18, then B is miniversal and so there is a map $\phi : A \to \Lambda$ inducing isomorphism $B\widehat{\otimes}_A\Lambda \simeq \widehat{\mathcal{O}}_{X,x}$. Setting $w_i = \phi(t_i)$ gives presentation (2.2). □

We are able to have both g_i and $\widetilde{g}_i$ polynomials due to $T^1(R)$ being finite-dimensional, so that $x^N = y^N = z^N = 0$ in $T^1(R)$ for $N \gg 0$. In fact we can describe algebraic deformations of the singularity up to henselization:

Corollary 2.21. *Suppose Λ is henselian and X, R are as above. In the notation of* Corollary 2.20 *there exist $w_i \in \mathfrak{m}_\Lambda$ so that we have an isomorphism of henselian local Λ-algebras*

$$\mathcal{O}^h_{X,x} \simeq \left(\frac{\Lambda[x,y,z]}{\widetilde{f} + \sum_{i=1}^r w_i \widetilde{g}_i} \right)^h_{(\mathfrak{m}_\Lambda, x, y, z)}$$

Proof. Write $R_{\mathrm{alg}} = (k[x,y,z]/f)^h_{(x,y,z)}$. By (2.2) and Artin approximation we have $\mathcal{O}^h_{X_k,x} \simeq R_{\mathrm{alg}}$. Consider the family

$$Y = \mathrm{Spec}\left(\frac{\Lambda[x,y,z,t_1,\cdots,t_r]}{\widetilde{f} + \sum_i t_i \widetilde{g}_i} \right) \to S \coloneqq \mathrm{Spec}(\Lambda[t_1,\cdots,t_r])$$

with points $s = (\mathfrak{m}_\Lambda, t_1, \cdots t_r)$ and $o = (\mathfrak{m}_\Lambda, x, y, z, t_1, \cdots, t_r)$. The family is flat near o and its completed local deformation over $\widehat{\mathcal{O}}_{S,s} = \widehat{\Lambda}[\![t_1, \cdots, t_r]\!]$ is miniversal by Proposition 2.18. By a theorem of Elkik[3] ([SGA 7_{II}, Exposé XV, §1.1.4]) $\mathrm{Spec}(\mathcal{O}^h_{Y,o}) \to \mathrm{Spec}(\mathcal{O}^h_{S,s})$ is a henselian miniversal deformation of R_{alg} and so by versality we get a local Λ-homomorphism $\phi : \mathcal{O}^h_{S,s} \to \Lambda$. Setting $w_i = \phi(t_i)$ completes the proof. □

2.4 Equivariant deformations of singularities. In the rest of the paper it will also be useful to enrich the deformations considered in Section 2.3 with a $\mathbb{G}_m$-equivariant structure, so we make this notion precise here. Once again, we fix a hypersurface singularity

$$R = \frac{k[\![x,y,z]\!]}{f(x,y,z)}$$

A $\mathbb{G}_m$-action on $A \in \mathrm{Comp}_{\Lambda,k}$ is a system of compatible algebraic $\mathbb{G}_m$-actions on $\mathrm{Spec}(A/\mathfrak{m}_A^n)$ for each $n \geq 1$, with trivial induced action on the residue field.

Definition 2.22. For $A \in \mathrm{Comp}_{\Lambda,k}$ equipped with a $\mathbb{G}_m$-action, an equivariant deformation of R over A is a deformation $B \in \mathrm{Alg}_A$ in the sense of Definition 2.11 equipped with a $\mathbb{G}_m$-action such that the structure map $A \to B$ and the fixed isomorphism $B/\mathfrak{m}_A B \xrightarrow{\sim} R$ are $\mathbb{G}_m$-equivariant.

There are equivariant notions of versality and miniversality which follow from Definitions 2.15 and 2.16 by declaring the lifting and the associated Kodaira–Spencer map to be compatible with the equipped $\mathbb{G}_m$-actions as above.

In this setting, $f(x,y,z)$ is quasihomogeneous with weights (w_x, w_y, w_z) and the natural $\mathbb{G}_m$-action on R is $(x,y,z) \mapsto (s^{w_x}x, s^{w_y}y, s^{w_z}z)$.

[3]Namely, henselian deformations of an isolated singularity over a henselian base are miniversal if and only if their induced formal deformations are miniversal.

Proposition 2.23 ([Slo80b], §2.5 (ii)). *Suppose $f(x,y,z)$ is quasihomogeneous with weights (w_x, w_y, w_z) and degree d. Then in the setting of* Proposition 2.18 *with $A = \Lambda[\![t_1, \cdots, t_r]\!]$, $\{g_i\}$ a homogeneous basis of $T^1(R)$ and $\widetilde{f}, \widetilde{g}_i$ homogeneous lifts to Λ, the formal local deformation*

$$B = \frac{A[\![x,y,z]\!]}{F(x,y,z)}, \quad F = \widetilde{f} + \sum_{i=1}^{r} t_i \widetilde{g}_i$$

is equivariantly miniversal with respect to the $\mathbb{G}_m$-action

$$s \cdot (x, y, z, t_1, \cdots, t_r) = (s^{w_x} x, s^{w_y} y, s^{w_z} z, s^{d_1} t_1, \cdots, s^{d_r} t_r), \quad d_i = d - \deg(g_i)$$

Note that taking weight components for the $\mathbb{G}_m$-action is an exact functor, so the argument of Slodowy here extends from k to Λ. For a description of the (doubled) weights w_x, w_y, w_z and d_i in the case of $f(x,y,z)$ parametrizing rational double points, see ([Slo80b, §7.4 Prop. 2]).

We also briefly discuss the problem of deforming minimal resolutions of a singularity. In what follows, let $\pi : X \to \mathrm{Spec}(R)$ be the minimal resolution of the isolated hypersurface singularity R with reduced exceptional divisor E, and set $\mathcal{X} = \widehat{X}_E$ for the formal completion of X along E; it is a smooth formal scheme with underlying space E.

Definition 2.24. Let $A \in \mathrm{Art}_{\Lambda,k}$. A *deformation of $\mathcal{X}$ over base A* is a flat morphism $\mathcal{X}_A \to \mathrm{Spec}(A)$ from a noetherian formal scheme, along with an isomorphism $\mathcal{X}_A \times_A \mathrm{Spec}(k) \simeq \mathcal{X}$ of formal schemes. We put

$$\mathrm{Def}_{\mathcal{X}}(A) = \{\text{deformations } \mathcal{X}_A \to \mathrm{Spec}(A)\}/\sim$$

Similarly, if $A \in \mathrm{Comp}_{\Lambda,k}$, then a deformation of $\mathcal{X}$ over A consists of a compatible system of deformations of $\mathcal{X}$ over $A_n := A/\mathfrak{m}^n A$.

Definitions 2.13, 2.15 and 2.16 extend naturally to the setting of deformations of formal schemes. Namely for $D = k[\varepsilon]/\varepsilon^2$ and $\mathcal{X}_A \in \mathrm{Def}_{\mathcal{X}}(A)$ we have a Kodaira–Spencer map

$$\mathrm{KS}_{\mathcal{X}_A} : \mathrm{T}_\Lambda(A) \to \mathrm{Def}_{\mathcal{X}}(D), \quad v \mapsto [\mathcal{X}_v]$$

where $\mathcal{X}_v$ is the pullback of $\mathcal{X}_A$ along $A \to D$. Then versality of a formal deformation is defined in terms of the usual lifting criterion, and miniversality is versality along with the Kodaira–Spencer map being an isomorphism.

Proposition 2.25. *When p is very good for the RDP singularity R, we have*

$$\mathrm{H}^1(\mathcal{X}, \mathcal{T}_{\mathcal{X}}) \simeq \bigoplus_{i \le r} \mathrm{H}^1(E_i, \mathcal{N}_{E_i/X}), \quad \mathrm{H}^2(\mathcal{X}, \mathcal{T}_{\mathcal{X}}) \simeq 0 \tag{2.3}$$

Proof. By [Hir19, Prop. 2.3], for $j_i : E_i \hookrightarrow \mathcal{X}$ we have an exact sequence

$$0 \to T_{\mathcal{X}}(-\log E) \to T_{\mathcal{X}} \to \bigoplus_{i \le r} (j_i)_* \mathcal{N}_{E_i/\mathcal{X}} \to 0 \tag{2.4}$$

with $\mathcal{N}_{E_i/\mathcal{X}} \simeq \mathcal{O}_{\mathbb{P}^1}(-2)$ as E_i are rational (-2)-curves (see Definition 2.1). Moreover $\mathrm{H}^2(\mathcal{X}, T_{\mathcal{X}}(-\log E)) = \mathrm{H}^2(\mathcal{X}, T_{\mathcal{X}}) = 0$ by Grothendieck vanishing. Thus taking cohomology in (2.4) gives $r \leq \dim \mathrm{H}^1(\mathcal{X}, T_X)$ and [Hir19, Prop. 2.2] gives $\dim \mathrm{H}^1(\mathcal{X}, T_{\mathcal{X}}) \leq \dim T^1(R) = r$ where $T^1(R)$ is the Tjurina algebra of the singularity (see Definition 2.13). We thus get both isomorphisms in (2.3). □

As a result we have that a deformation of $\mathcal{X}$ over $\Lambda[\![s_1, \cdots, s_r]\!]$ is miniversal if and only if its associated Kodaira–Spencer map is an isomorphism.

2.5 Artin components. We briefly discuss special components of the deformation space when R is the completed local ring of an arbitrary rational surface singularity.

Let (U, x) be an affine open neighborhood of the singularity R in that $R = \widehat{\mathcal{O}}_{U,x}$. Isolated singularities admit formal miniversal deformations over base Λ, and in particular such deformations have an algebraic versal representative ([Elk73, Part IV, Thms. 7, 8]). Thus we may choose a flat family $f : \mathcal{U} \to S$ of finite-type Λ-schemes, with relative singular locus finite over S, such that $\mathcal{U}_s \simeq U$, $A \coloneqq \widehat{\mathcal{O}}_{S,s} \in \mathrm{Comp}_{\Lambda,k}$ and $B \coloneqq \widehat{\mathcal{O}}_{\mathcal{U},x} \in \mathrm{Def}_R(A)$ is a versal deformation of R (see Definitions 2.11 and 2.16 (i)). After pullback to $\mathrm{Spec}(A)$, morphism f is an algebraic realization of B in the sense of Definition 2.11 (ii).

Definition 2.26 ([Art74], §1)**.** The simultaneous resolution functor

$$\mathrm{Res}_{\mathcal{U}/S} : (\mathrm{Sch}/S)^{\mathrm{op}} \to \mathrm{Set}, \quad \mathrm{Res}_{\mathcal{U}/S}(T) = \{\pi : \mathcal{X} \to U \times_S T\}/\simeq$$

parametrizes simultaneous resolutions of the family $\mathcal{U} \to S$ in the sense that $\mathcal{X}$ is an algebraic space smooth over T and π is a simultaneous resolution (Definition 2.4).

By ([Art74, Thm. 1]), $\mathrm{Res}_{\mathcal{U}/S}$ is representable by an algebraic space $\mathcal{R}$ over S. Let $a \in \mathcal{R}(k)$ correspond to the minimal resolution $\pi : \widetilde{U} \to U$ of the rational singularity R and set $C \coloneqq \widehat{\mathcal{O}}_{\mathcal{R},a}$.

Definition 2.27. The *Artin component* of the versal deformation $\mathcal{U} \to S$ of R is the scheme-theoretic image of $\mathrm{Spf}(C) \to \mathrm{Spf}(A) = \mathrm{Spf}(\widehat{\mathcal{O}}_{S,s})$. Equivalently it is $\mathrm{Spf}(A/I)$ where $I = \ker(A \to C)$ is induced from the structure map.

Proposition 2.28. *Assume R is a rational singularity and let $f : \mathcal{U} \to S$, A and C be as above. Then C is a finite A-algebra and a formally smooth Λ-algebra, the Artin component is an irreducible component of* $\mathrm{Spf(A)}$ *and the induced family $f|_{A/I}$ over* $\mathrm{Spec}(A/I)$ *admits a simultaneous resolution after finite base-change* $\mathrm{Spec}(C) \to \mathrm{Spec}(A/I)$.

Proof. By ([Art74, Thm. 3]) $\mathrm{Spf}(C) \to \mathrm{Spf}(A)$ is a finite map onto an irreducible component of $\mathrm{Spf}(A)$ i.e. its image is $\mathrm{Spf}(A/I)$ with I a minimal prime. Since $\mathrm{H}^2(\widetilde{U}, T_{\widetilde{U}}) = 0$ for the minimal resolution of the rational singularity of U, deformations of $\widetilde{U}$ over Λ are unobstructed and so C is a formally smooth Λ-algebra by ([Art74, Lem. 3.3]). By construction $\mathrm{Spec}(C) \to \mathcal{R}$, so the representing property of $\mathcal{R}$ yields a simultaneous resolution of the base-changed family $\mathcal{U} \times_S \mathrm{Spec}(C)$. □

Remark 2.29. When R is a rational double point (i.e. a rational hypersurface singularity), its miniversal base A is a domain by Proposition 2.18 so minimal primes are zero and the Artin component equals $\mathrm{Spf}(A)$. We will see that versal deformations of rational double points admit a simultaneous resolution after completion and base-change, without shrinking the base (Proposition 4.5).

3. The geometry of the Grothendieck alteration

We now collect the necessary Lie-theoretic prerequisites in order to define miniversal deformations of rational double points in terms of the adjoint quotient of Lie algebras. The main goal is to define and study the classical Grothendieck–Springer resolution over mixed characteristic integral bases (Section 3.4). To this end, we first extend the relevant notions of nilpotent, semisimple and (sub)regular elements to the mixed characteristic setting and construct an integral version of the adjoint quotient in preparation for Theorem 3.8.

3.1 Notations and conventions. We work over a mixed characteristic complete discrete valuation ring Λ with algebraically closed residue field k; as a trait, its closed point is denoted s and its generic point is denoted η.

Fix a split, simply connected, simple group G/Λ along with a choice of split maximal torus and Borel $T \subset B \subset G$, unipotent radical $U = \mathcal{R}_u(B)$ so that $B = U \rtimes T$, and set

$$\mathfrak{g} = \mathrm{Lie}(G),\ \mathfrak{h} = \mathrm{Lie}(T), \mathfrak{b} = \mathrm{Lie}(B),\ \mathfrak{n} = \mathrm{Lie}(U)$$

Write $r := \mathrm{rk}(G) = \mathrm{rk}_\Lambda(\mathfrak{h})$ for the rank of Lie algebra $\mathfrak{g}$[4]. Identification $B = U \rtimes T$ yields a decomposition of Λ-modules

$$\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n},\quad \mathfrak{b}/\mathfrak{n} \simeq \mathfrak{h} \tag{3.1}$$

We will sometimes identify $\mathfrak{g}$ with its affine scheme $\mathrm{Spec}(\mathrm{Sym}_\Lambda(\mathfrak{g}^\vee))$, so that for $R \in \mathrm{Alg}_\Lambda$ its R-points are $R \otimes_\Lambda \mathfrak{g}$ and its coordinate ring is $\Lambda[\mathfrak{g}]$; similar notations will be used for the Lie subalgebras of $\mathfrak{g}$.

Write $W = N_G(T)/T$ for the Weyl group of (G,T); it is a finite étale (in fact constant) group scheme. It comes naturally equipped with an adjoint action on $\mathfrak{h}$ via $\mathrm{Ad} : G \to \mathrm{Aut}_\Lambda(\mathfrak{g})$; a priori it is an action of $N_G(T)$ but it descends to W as T acts trivially on $\mathfrak{h}$.

Let $\Phi = \Phi(G,T)$ be the associated root lattice of (G,T). We make the following note once and for all:

Assumption 3.1. G is of ADE type so that Φ is a semisimple simply-laced root lattice, and residue characteristic $p = \mathrm{char}(k)$ is *very good* for Φ (see Definition 2.3).

Setting $\Phi^\vee = \mathrm{Hom}_\mathbb{Z}(\Phi, \mathbb{Z})$ we may identify it with the character lattice $X^*(T)$ via Assumption 3.1, and moreover $\Phi \otimes \mathbb{Q}_\ell \simeq \Phi^\vee \otimes \mathbb{Q}_\ell$ in good characteristic; the restrictions on p ensure G/Λ is root-smooth ([BČ22, §4.1.1]) and p is a non-torsion prime for the root datum of (G,T) ([BČ22, §4.1.12]). We remark that p good is not a sufficient restriction as e.g. $G = \mathrm{SL}_2$ is not root-smooth when $p = 2$.

[4]Not to be confused with $\mathrm{rk}_\Lambda(\mathfrak{g})$, which is the dimension of G/Λ.

3.2 Nilpotent and regular loci. We retain the assumptions of Section 3.1. Over the generic point $\eta \to \mathrm{Spec}(\Lambda)$ we have the usual reduced nilpotent variety $\mathcal{N}_\eta \subset \mathfrak{g}_\eta$, where $\mathfrak{g}_\eta$ is a characteristic zero Lie $k(\eta)$-algebra.

Definition 3.2. The nilpotent scheme $\mathcal{N} \subset \mathfrak{g}$ is the schematic closure $\overline{\mathcal{N}_\eta}^{\,\mathrm{sch}}$ of $\mathcal{N}_\eta$ inside $\mathfrak{g}$. If $R = \mathrm{Sym}_\Lambda(\mathfrak{g}^\vee)$ and $I \subset R \otimes_\Lambda K$ is the ideal of $\mathcal{N}_\eta$, then $\mathcal{N} = \mathrm{Spec}(R/I \cap R)$.

The nilpotent scheme is Λ-flat and, under our very good assumptions on p, its special fiber $\mathcal{N}_s$ is the usual normal nilpotent variety of $\mathfrak{g}_s$ ([Cot26, Thm.4.12]). Note for $x \in \mathfrak{g}(\Lambda)$, we have $x \in \mathcal{N}(\Lambda)$ if and only if $x_{\overline{\eta}} \in \mathcal{N}(\overline{k(\eta)})$; one direction follows from $\Lambda \hookrightarrow \overline{k(\eta)}$ while the converse follows by base-change.

For the next set of definitions, we recall that for a geometric point $\bar{t} \to \mathrm{Spec}(\Lambda)$ (i.e. the closed point s and the geometric generic point $\overline{\eta}$), the centralizer of an element $x \in \mathfrak{g}_{\bar{t}}(k(\bar{t}))$ under the adjoint action is denoted $C_{G_{\bar{t}}}(x)$.

Definition 3.3. Let r be the rank of Lie algebra $\mathfrak{g}/\Lambda$. A section $x \in \mathfrak{g}(\Lambda)$ will be called:

(i) *regular* if $\dim C_{G_{\bar{t}}}(x_{\bar{t}}) = r$ for all $\bar{t} \to \mathrm{Spec}(\Lambda)$ ([BČ22, §4.2.1]).

(ii) *subregular nilpotent*[5] if $\dim C_{G_{\bar{t}}}(x_{\bar{t}}) = r + 2$ for all $\bar{t} \to \mathrm{Spec}(\Lambda)$ and $x \in \mathcal{N}(\Lambda)$ ([Slo80b, §5.4]).

(iii) *regular semisimple* if $C_{G_{\bar{t}}}(x_{\bar{t}})$ is a maximal torus in $G_{\bar{t}}$ for all $\bar{t} \to \mathrm{Spec}(\Lambda)$ ([BČ22, Lemma 4.2.7]).

Since the residue characteristic is very good for $\Phi(G,T)$ (in particular G/Λ is root-smooth), the loci of regular semisimple and regular sections in $\mathfrak{g}(\Lambda)$ form respectively the Λ-points of G-stable, $\mathrm{Spec}(\Lambda)$-fiberwise dense open subschemes $\mathfrak{g}^{\mathrm{rs}} \subset \mathfrak{g}^{\mathrm{reg}} \subset \mathfrak{g}$ whose formation commutes with base-change (cf. [BČ22, §4.1.5, §4.2.1-2]). We can accordingly say $x \in \mathfrak{g}(\Lambda)$ is a regular semisimple (resp. regular) section if and only if $x \in \mathfrak{g}^{\mathrm{rs}}(\Lambda)$ (resp. $x \in \mathfrak{g}^{\mathrm{reg}}(\Lambda)$).

We also define locus $\mathfrak{h}^{\mathrm{rs}} := \mathfrak{h} \cap \mathfrak{g}^{\mathrm{rs}}$. For a root $\alpha : T \to \mathbb{G}_m$ we have $d\alpha : \mathfrak{h} \to \Lambda$ i.e. $d\alpha \in \mathfrak{h}^\vee$. By [BČ22, §4.1.5], one can view $\mathfrak{h}^{\mathrm{rs}}$ as the complement of root hyperplanes in $\mathfrak{h}$, i.e. ranging over all roots $\alpha \in \Phi$ we have

$$\mathfrak{h}^{\mathrm{rs}} = \mathfrak{h} \setminus \cup_{\alpha \text{ root}} V(d\alpha)$$

Example 3.4. As already remarked, for an integral section $x \in \mathfrak{g}(\Lambda)$, it suffices to have the generic fiber $x_{\overline{\eta}} \in \mathcal{N}(k(\overline{\eta}))$ in order to get $x \in \mathcal{N}(\Lambda)$ because $\mathcal{N}$ is a closed subscheme. On the other hand, as $\mathfrak{g}^{\mathrm{rs}}, \mathfrak{g}^{\mathrm{reg}}$ are open, x_s regular (resp. regular semisimple) implies $x_{\overline{\eta}}$ regular (resp. regular semisimple). The converse implications are generally false, while for subregular nilpotent elements neither implication holds and we require the fiberwise constancy of $\dim C_{G_{\bar{t}}}(x_{\bar{t}})$: For $x = E_{12} + pE_{23} \in \mathfrak{sl}_3(\Lambda)$, x_s is subregular nilpotent but $x_{\overline{\eta}}$ is regular, and for $y = pE_{12}$, $y_{\overline{\eta}}$ is subregular nilpotent but $y_s = 0$ is not subregular.

[5] Of course one can omit $x \in \mathcal{N}(\Lambda)$ and define subregular sections in general, but we will only work with subregular nilpotent elements. Such subregular sections always exist by [McN08, §5.4, pp. 159-160].

More broadly, since G/Λ is split, G_s and $G_{\bar{\eta}}$ have the same root datum and so since p is (very) good, the two sets of nilpotent orbits in $\mathcal{N}(k(\overline{\eta})), \mathcal{N}(k)$ (i.e. the orbits under the adjoint G-action) have the same classification. Then purity of a section $x \in \mathcal{N}(\Lambda)$ in the sense that $\dim C_{G_{\bar{t}}}(x_{\bar{t}})$ is constant for $\bar{t} \to \mathrm{Spec}(\Lambda)$ ensures $x_{\bar{t}}$ lie in the same type of nilpotent orbit ([Cot22, Lemma 5.7]).

3.3 The adjoint quotient. We turn to explaining the construction and properties of the adjoint quotient for algebraic groups over Λ. A classical comprehensive reference for the usual construction over algebraically closed fields (including positive characteristic) is [Slo80b, §3.10-3.14].

The adjoint action of G on $\mathfrak{g}$ and the induced adjoint action of W on $\mathfrak{h}$ (see Section 3.1) yield categorical quotient maps

$$\chi : \mathfrak{g} \longrightarrow \mathfrak{g}/\!\!/G := \mathrm{Spec}\left(\Lambda[\mathfrak{g}]^G\right), \quad \psi : \mathfrak{h} \longrightarrow \mathfrak{h}/\!\!/W := \mathrm{Spec}\left(\Lambda[\mathfrak{h}]^W\right)$$

We call χ the *adjoint quotient*; ψ is a finite branched cover. Restricting invariant functions along $\mathfrak{h} \subset \mathfrak{g}$ yields in addition a natural morphism $\rho : \mathfrak{h}/\!\!/W \to \mathfrak{g}/\!\!/G$.

Theorem 3.5. *With the above notations and the hypotheses of* Section 3.1*:*

(i) *Map ρ is an isomorphism $\mathfrak{h}/\!\!/W \simeq \mathfrak{g}/\!\!/G \simeq \mathbb{A}^r_\Lambda$, where the latter isomorphism is non-canonical. Moreover formation of both quotients commutes with base-change.*

(ii) *The adjoint quotient χ is faithfully flat with geometrically normal fibers of dimension $\dim(G_s) - r$ and the restriction $\chi^{reg} : \mathfrak{g}^{reg} \to \mathfrak{h}/\!\!/W$ is smooth surjective.*

(iii) *We have $\mathcal{N} \simeq \chi^{-1}(0)$ where $0 \in (\mathfrak{h}/\!\!/W)(\Lambda)$ is the zero section.*

(iv) *Map ψ is finite flat, locally free of degree $|W|$. Denoting $(\mathfrak{h}/\!\!/W)^{rs} = \psi(\mathfrak{h}^{rs})$ we have $\chi^{-1}((\mathfrak{h}/\!\!/W)^{rs}) = \mathfrak{g}^{rs}$ and $\psi^{rs} : \mathfrak{h}^{rs} \to (\mathfrak{h}/\!\!/W)^{rs}$ is a finite étale W-torsor.*

Proof. (i) is [BČ22, Thm. 4.1.10, Prop. 4.1.14, Rem. 4.1.15], the non-canonical isomorphism holds because G/Λ is split. (ii) is [BČ22, Cor. 4.1.18, Prop. 4.2.6] and normality of the fibers follows from [Slo80b, §3.10 Thm. (v), §3.14]. For (iii), note that $\chi^{-1}(0)$ is Λ-flat by (ii) and generic fiber $\chi^{-1}(0)_{\overline{\eta}}$ is $\mathcal{N}_\eta$ by construction. As flat closed schemes over a discrete valuation ring are the schematic closure of their generic fiber we have $\mathcal{N} \simeq \chi^{-1}(0)$. (iv) is [BČ22, Cor. 4.1.18, Lem. 4.2.12] as geometric points in $\mathfrak{g}^{\mathrm{rs}}$ are conjugate to elements in $\mathfrak{h}^{\mathrm{rs}}$; the torsor is finite étale because W is a finite étale Λ-scheme. □

Remark 3.6. Over algebraically closed fields, the adjoint quotient maps x to the conjugacy class of its semisimple part via the Jordan decomposition. This interpretation however does not hold over Λ as there need not be a Jordan decomposition specializing to the Jordan decompositions in the fibers.

3.4 Relative Grothendieck–Springer resolutions. In this section we aim to construct an analogue of the Grothendieck–Springer resolution over Λ. As before, a classical reference for the construction over algebraically closed fields is [Slo80b, §4.4, §4.7].

Recall that the free module $\mathfrak{b}$ is a B-representation through the adjoint action of B on $\mathfrak{b}$ (see §3.1), and so via the B-torsor $G \to G/B$ we can construct the associated *adjoint* bundle

$$\widetilde{\mathfrak{g}} := G \times^B \mathfrak{b} := (G \times_\Lambda \mathfrak{b})/B, \quad b \cdot (g, x) = (gb^{-1}, \mathrm{Ad}_b(x)) \text{ for } b \in B, g \in G, x \in \mathfrak{b} \tag{3.2}$$

with the induced B-action on $G \times_\Lambda \mathfrak{b}$ indicated on the right. Note we have a closed immersion $\widetilde{g} \hookrightarrow G \times^B \mathfrak{g}$ and the latter is identified with $G/B \times_\Lambda \mathfrak{g}$ via $[g, x] \mapsto (gB, \mathrm{Ad}(g)x)$. Together with the B-invariant projection $G \times_\Lambda \mathfrak{b} \to G/B$ we obtain two maps

$$\begin{array}{ccc} \widetilde{\mathfrak{g}} & \xrightarrow{\pi} & \mathfrak{g} \\ \downarrow{\scriptstyle p} & & \\ G/B & & \end{array}$$

where p is the Zariski-locally trivial vector bundle associated with the B-representation $\mathfrak{b}$ by descent along B-torsor $G \to G/B$, and π is the projection $\widetilde{\mathfrak{g}} \hookrightarrow G/B \times_\Lambda \mathfrak{g} \to \mathfrak{g}$, sending $[g, x] \mapsto \mathrm{Ad}(g)x$. We call projective morphism π the *Grothendieck alteration*.

Remark 3.7. An equivalent description is $\widetilde{\mathfrak{g}} = \{(B', x) \in G/B \times_\Lambda \mathfrak{g} \mid x \in \mathrm{Lie}(B')\}$. Moreover π is dominant and the formation of $\widetilde{\mathfrak{g}}$ commutes with base-change on Λ; see [BČ22, §4.1.7] for details.

We next construct a morphism $\widetilde{\chi} : \widetilde{\mathfrak{g}} \to \mathfrak{h}$. The quotient map $B \to T \simeq B/U$ yields an exact sequence of B-modules

$$0 \to \mathfrak{n} \to \mathfrak{b} \to \mathfrak{h} \to 0$$

and since B acts trivially on $\mathfrak{h}$ (as its conjugation action on B/U is trivial), we get an induced exact sequence of vector bundles

$$0 \to G \times^B \mathfrak{n} \to \widetilde{\mathfrak{g}} \to G \times^B \mathfrak{h} \simeq G/B \times_\Lambda \mathfrak{h} \to 0 \tag{3.3}$$

Then we define $\widetilde{\chi} : \widetilde{\mathfrak{g}} \to G/B \times_\Lambda \mathfrak{h} \to \mathfrak{h}$ by sending $[g, x] \mapsto x \mod \mathfrak{n}$.

Theorem 3.8. *With the above notations, we have a commutative diagram*

$$\begin{array}{ccc} \widetilde{\mathfrak{g}} & \xrightarrow{\pi} & \mathfrak{g} \\ \downarrow{\scriptstyle \widetilde{\chi}} & & \downarrow{\scriptstyle \chi} \\ \mathfrak{h} & \xrightarrow{\psi} & \mathfrak{h}/\!\!/W \end{array} \tag{3.4}$$

whose construction commutes with base-change on Λ. Moreover:

(i) *$\widetilde{\chi}$ is smooth surjective and π is finite locally free of degree $|W|$ over $\mathfrak{g}^{reg}$. Moreover, the diagram induces a map $q : \widetilde{\mathfrak{g}} \to \mathfrak{g} \times_{\mathfrak{h}/\!\!/W} \mathfrak{h}$ which is projective birational.*

(ii) *Restricting to regular semisimple loci* $\widetilde{\mathfrak{g}}^{rs} := \pi^{-1}(\mathfrak{g}^{rs}) \to \mathfrak{g}^{rs}$ *yields a cartesian diagram*

$$\begin{array}{ccc} \widetilde{\mathfrak{g}}^{rs} & \xrightarrow{\pi^{rs}} & \mathfrak{g}^{rs} \\ \downarrow{\scriptstyle \widetilde{\chi}^{rs}} & & \downarrow{\scriptstyle \chi^{rs}} \\ \mathfrak{h}^{rs} & \xrightarrow{\psi^{rs}} & (\mathfrak{h}/\!\!/W)^{rs} \end{array} \tag{3.5}$$

where π^{rs}, ψ^{rs} *are finite étale* W*-torsors; here* $(-)^{rs}$ *denotes the restriction of the relevant maps to the regular semisimple loci.*

(iii) *The map* $G/T \times_\Lambda \mathfrak{h}^{rs} \to \widetilde{\mathfrak{g}}^{rs}$, $(gT, h) \mapsto [g, h]$ *is an isomorphism and the map* $\widetilde{\chi}^{rs}$ *is* W*-equivariant with respect to the following action on the source:*

$$a_w(gT, h) = (g\widetilde{w}^{-1}T, Ad(\widetilde{w})h), \quad \widetilde{w} \in N_G(T) \text{ lift of } w \in W$$

(iv) *Base-changing diagram* (3.4) *along any geometric point* $\bar{t} \to \mathrm{Spec}(\Lambda)$ *yields a simultaneous resolution diagram in the sense that for all* $h \in \mathfrak{h}_{\bar{t}}(k(\bar{t}))$ *the induced map*

$$\pi(\bar{t}, h) : \widetilde{\chi}_{\bar{t}}^{-1}(h) \to \chi_{\bar{t}}^{-1}(\psi_{\bar{t}}(h)) \tag{3.6}$$

is a resolution of singularities. Here $(-)_{\bar{t}}$ *denotes the relevant pulled-back maps and spaces from* (3.4) *along* $\bar{t} \to \mathrm{Spec}(\Lambda)$.

Proof. Commutativity and base-change compatibility of (3.4) follow from [BČ22, §4.1.7, (4.1.7.4)]. For (i), $\widetilde{\chi}$ is a composition of smooth surjections, the first map being locally a projection of vector bundles, hence it is smooth surjective. By [BČ22, Lem. 4.2.12] we have $\widetilde{\mathfrak{g}}^{\mathrm{reg}} := \pi^{-1}(\mathfrak{g}^{\mathrm{reg}}) \simeq \mathfrak{g}^{\mathrm{reg}} \times_{\mathfrak{h}/\!\!/W} \mathfrak{h}$ and $\pi|_{\widetilde{\mathfrak{g}}^{\mathrm{reg}}}$ is the pullback of ψ which is finite locally free of degree $|W|$ by Theorem 3.5(iv), so we get the same property for π over $\mathfrak{g}^{\mathrm{reg}}$. For the "moreover" part of (i), q is induced by the universal property of fiber products and factors as a closed immersion and $\widetilde{\mathfrak{g}} \times_{\mathfrak{h}/\!\!/W} \mathfrak{h} \to \mathfrak{g} \times_{\mathfrak{h}/\!\!/W} \mathfrak{h}$ which is a base-change of π, hence q is projective and an isomorphism over open dense locus $\widetilde{\mathfrak{g}}^{\mathrm{reg}}$, i.e. it is also birational.

For (ii), by [BČ22, Lem. 4.2.12] we have an identification

$$\widetilde{\mathfrak{g}}^{\mathrm{rs}} = \widetilde{\mathfrak{g}}^{\mathrm{reg}} \times_{\mathfrak{g}^{\mathrm{reg}}} \mathfrak{g}^{\mathrm{rs}} \simeq \mathfrak{h}^{\mathrm{rs}} \times_{(\mathfrak{h}/\!\!/W)^{\mathrm{rs}}} \mathfrak{g}^{\mathrm{rs}}$$

so (3.5) is a cartesian diagram and $\pi^{\mathrm{rs}}, \psi^{\mathrm{rs}}$ are finite étale W-torsors by [BČ22, Prop. 4.1.8]. (iii) also follows from [BČ22, Prop. 4.1.8, (4.1.8.1)]. Finally (iv) is the content of the Grothendieck–Springer simultaneous resolution theorem over algebraically closed fields of very good characteristic, which is proven in [Slo80b, §4.7]; the normality of the target of $\pi(\bar{t}, h)$ is supplied by Theorem 3.5(ii). □

Remark 3.9. While $\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ is commonly referred to as the Grothendieck–Springer resolution, we insist on calling π the Grothendieck alteration as it is a map generically of degree $|W|$ and thus not birational. The factoring map $q : \widetilde{\mathfrak{g}} \to \mathfrak{g} \times_{\mathfrak{h}/\!\!/W} \mathfrak{h}$ is proper birational and furnishes the simultaneous resolution of $\chi : \mathfrak{g} \to \mathfrak{h}/\!\!/W$ after base-change along ψ.

For later purposes, we also record here a version of the Grothendieck–Springer resolution when the fibers are Λ-schemes.

Corollary 3.10. *Let $x \in \mathfrak{h}(\Lambda)$ be an integral section, viewed as a map $x : \mathrm{Spec}(\Lambda) \to \mathfrak{h}$. The induced map*

$$\pi(x) : \widetilde{\mathfrak{g}} \times_{\mathfrak{h}} \mathrm{Spec}(\Lambda) \to \mathfrak{g} \times_{\mathfrak{h}/\!\!/W} \mathrm{Spec}(\Lambda)$$

is a resolution of singularities of Λ-schemes so that the Λ-fiberwise induced maps are (3.6). Here the base-changes to $\mathrm{Spec}(\Lambda)$ are understood to be via maps x on the source and $\psi(x)$ on the target.

Proof. Note the source and target of $\pi(x)$ are closed Λ-subschemes of $\widetilde{\mathfrak{g}}$ and $\mathfrak{g}$, respectively. Then $\widetilde{\mathfrak{g}} \times_{\mathfrak{h}} \mathrm{Spec}(\Lambda)$ is smooth over $\mathrm{Spec}(\Lambda)$ by base-change from $\widetilde{\chi}$ (Theorem 3.8(i)), $\pi(x)$ is the base-change of $q : \widetilde{\mathfrak{g}} \to \mathfrak{g} \times_{\mathfrak{h}/\!\!/W} \mathfrak{h}$ hence it is projective, and it is an isomorphism over $(\mathfrak{g} \times_{\mathfrak{h}/\!\!/W} \mathrm{Spec}(\Lambda)) \cap \mathfrak{g}^{\mathrm{reg}}$, so it is also birational (Theorem 3.8(i) again) and $\pi(x)$ is a resolution. Lastly, base-change along $\bar{t} \to \mathrm{Spec}(\Lambda)$ yields

$$\pi(x)_{\bar{t}} : \widetilde{\chi}_{\bar{t}}^{-1}(x_{\bar{t}}) \to \chi_{\bar{t}}^{-1}(\psi_{\bar{t}}(x_{\bar{t}}))$$

and so $\pi(x)_{\bar{t}}$ is resolution $\pi(\bar{t}, x_{\bar{t}})$ in the notation of Theorem 3.8(iv). □

In particular, choosing the zero section $x = 0 \in \mathfrak{h}(\Lambda)$ in Corollary 3.10 yields the relative Springer resolution $G \times^B \mathfrak{n} \to \mathcal{N}$; further base-change to a geometric point $\bar{t} \to \mathrm{Spec}(\Lambda)$ yields the classical Springer resolution over fields (see [Slo80b, §4.1] for the unipotent version).

4. Integral transverse slices

The purpose of this section is to define a homogeneous transverse slice to subregular nilpotent sections, whose construction works in all very good characteristics. The end results are Proposition 4.5 and Theorem 4.7, which use the geometry of the Grothendieck alteration to show that restricting the adjoint quotient to this slice yields an equivariantly miniversal deformation of a rational double point singularity, and the deformations of its minimal resolution are also captured by this construction.

4.1 Homogeneous transverse slices. Throughout we work over a mixed characteristic complete discrete valuation ring Λ with algebraically closed residue field k. Let G/Λ be a split simple simply-connected group of ADE type, $\mathfrak{g} = \mathrm{Lie}(G)$ its Lie algebra of rank r and assume the residue characteristic p is very good (see Definition 2.3) for the associated root lattice Φ. We view $\mathfrak{g}$ both as a finite free Λ-module and as its associated affine Λ-scheme.

Definition 4.1. Fix a subregular nilpotent section $x \in \mathfrak{g}(\Lambda)$ as in Definition 3.3 (ii). For a smooth locally closed Λ-subscheme $\Sigma \subset \mathfrak{g}$ containing section x, we say Σ is a *relative transverse slice* at x if for every geometric point $\bar{t} \to \mathrm{Spec}(\Lambda)$ we have decomposition

$$T_{x_{\bar{t}}}\mathfrak{g}_{\bar{t}} = [\mathfrak{g}_{\bar{t}}, x_{\bar{t}}] \oplus T_{x_{\bar{t}}}\Sigma_{\bar{t}}$$

Here $T_{x_{\bar{t}}}\Sigma_{\bar{t}} = \mathrm{Hom}_{k(t)}(x_{\bar{t}}^*\Omega^1_{\Sigma/\Lambda}, \overline{k(t)})$. We note that the associated adjoint action map $G \times \Sigma \to \mathfrak{g}$ is smooth in a neighborhood of $G \times \{x\}$.

By a simple base-change argument we have the following preliminary lemma:

Lemma 4.2. *Let $f : X \to \mathfrak{g}$ be a G-equivariant morphism of Λ-schemes and set $\Sigma_X := X \times_{\mathfrak{g}} \Sigma$. Then we have a cartesian diagram*

$$\begin{array}{ccc} G \times \Sigma_X & \longrightarrow & X \\ \downarrow & & \downarrow f \\ G \times \Sigma & \longrightarrow & \mathfrak{g} \end{array}$$

and so the action map $G \times \Sigma_X \to X$ is smooth if the action map $G \times \Sigma \to \mathfrak{g}$ is smooth.

Since $x \in \mathfrak{g}(\Lambda)$ is fiberwise subregular and Λ is strictly henselian, we can associate to x a cocharacter $\lambda : \mathbb{G}_m \to G$ such that $\lambda_{\bar{t}}$ is a cocharacter associated to $x_{\bar{t}}$ (see [Cot22, Def. 2.9, Lemma 5.7 (3)]). In other words we have a $\mathbb{G}_m$-action on $\mathfrak{g}$ and a compatible $\mathbb{Z}$-grading

$$\begin{gathered} \mathrm{Ad}_{\lambda(t)}x = t^2 x, \quad t \in \mathbb{G}_m(\Lambda) \\ \mathfrak{g} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}(i), \quad \mathfrak{g}(i) := \{y \in \mathfrak{g} \mid \mathrm{Ad}_{\lambda(t)}y = t^i y\} \end{gathered} \tag{4.1}$$

Proposition 4.3. *There is a graded free summand $\mathfrak{a} \subset \bigoplus_{i \leq 0} \mathfrak{g}(i)$ such that $\mathfrak{g} = [x, \mathfrak{g}] \oplus \mathfrak{a}$ and the affine scheme $\Sigma = x + \mathfrak{a}$ is a relative transverse slice of (relative) dimension $r+2$ with smooth action map $G \times \Sigma \to \mathfrak{g}$.*

Proof. Since we are in very good characteristic, the orbit maps for the adjoint action on $\mathfrak{g}$ are separable ([Slo80b, §5.1, Rem. (2)]) and so $\dim \ker(\mathrm{ad}(x_{\bar{t}})) = r + 2$ on geometric fibers, by subregularity. Thus $M = \mathrm{coker}(\mathrm{ad}(x))$ is a free Λ-module of rank $r+2$; we may write $M = \mathfrak{g}/[x, \mathfrak{g}]$.

Now $x \in \mathfrak{g}(2)$ so $[x, \mathfrak{g}(i)] \subseteq \mathfrak{g}(i+2)$ and we have a grading on $M = \bigoplus_i M(i)$ with $M(i) = \mathfrak{g}(i)/[x, \mathfrak{g}(i-2)]$ also a free Λ-module (as a summand). By passing to the characteristic-zero geometric generic fiber $\bar{t} = \mathrm{Spec}(\overline{K}) \to \Lambda$ we have that $M(i) \otimes_\Lambda \overline{K} = 0$ for $i \geq 1$ since $\mathfrak{g}_{\bar{t}}$ is a completely reducible $\mathfrak{sl}_2$-representation in characteristic zero (see e.g. [Slo80b, §7.4]), thus $M(i) = 0$ for $i \geq 1$. By choosing a (graded) splitting of $\mathfrak{g}(i) \twoheadrightarrow M(i)$ and denoting its image by $\mathfrak{a}(i)$, we set

$$\mathfrak{a} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{a}(i) = \bigoplus_{i \leq 0} \mathfrak{a}(i), \quad \mathfrak{a}(i) \simeq M(i)$$

and by construction $\mathfrak{g}(i) = [x, \mathfrak{g}(i-2)] \oplus \mathfrak{a}(i)$ so $\mathfrak{g} = [x, \mathfrak{g}] \oplus \mathfrak{a}$ and $\Sigma = x + \mathfrak{a}$ is a relative transverse slice at x.

It remains to check that the action map is smooth everywhere. Define a $\mathbb{G}_m$-action on $\mathfrak{g}$ via

$$\mu_t(y) = t^2 \mathrm{Ad}_{\lambda(t)^{-1}} y, \quad t \in \mathbb{G}_m(\Lambda) \tag{4.2}$$

By construction it preserves Σ, fixes x and the weights on each nontrivial $\mathfrak{a}(i)$ are $2-i>0$ so the action contracts Σ to x; concretely, this means that $\mu:\mathbb{G}_m\times_\Lambda\Sigma\to\Sigma$ extends to an action $\mathbb{A}^1_\Lambda\times_\Lambda\Sigma\to\Sigma$ and $\mu_0(y)=x$. The locus of Σ where the action map at $(1,y)\in G\times\Sigma$ is not smooth is closed and stable under this contracting action, hence must contain x, contradicting local smoothness at x. Translation by G makes $G\times\Sigma\to\mathfrak{g}$ fiberwise smooth, hence smooth as $G\times\Sigma$ is flat over Λ. □

Remark 4.4. If we replace Λ by an algebraically closed field of characteristic zero or $p>4\mathrm{Cox}(\mathfrak{g})+2$ (where $\mathrm{Cox}(\mathfrak{g})$ denotes the Coxeter number of the simple Lie algebra $\mathfrak{g}$), Σ can be directly constructed via an $\mathfrak{sl}_2$-triple (x,y,h) and $\mathfrak{a}=\ker(\mathrm{ad}y)$; Σ is known as a Slodowy slice (cf. [Slo80b, §7.4]). Proposition 4.3 gives essentially a relative-to-Λ version of the construction in [Spa84].

We now make precise the sense in which Σ is *homogeneous*. We call a regular function $f\in\Lambda[\Sigma]$ homogeneous of degree m under action $\mu:\mathbb{G}_m\times\Sigma\to\Sigma$ if

$$\mu^*f=t^m\otimes f\in\Lambda[t^\pm]\otimes_\Lambda\Lambda[\Sigma]$$

This notion makes sense for regular functions on any affine scheme with a $\mathbb{G}_m$-action. For Σ specifically write $\Lambda[\Sigma]=\Lambda[u_1,\cdots,u_{r+2}]$ where u_i are coordinate functions on Σ; for a homogeneous Λ-basis $\{e_j\in\mathfrak{a}(i_j)\}$ of $\mathfrak{a}$ we have $u_i(x+\sum_j a_je_j)=a_i$. One directly checks that the $\mathbb{G}_m$-action on Σ defined in (4.2) yields u_j of weight $2-i_j>0$; here, i_j is the degree of element e_j under the $\mathbb{Z}$-grading in (4.1).

4.2 Grothendieck alterations for transverse slices. We recall the setting of Section 3.1, and in particular retain Assumption 3.1 for G and root lattice $\Phi(G,T)$. Setting $\mathfrak{g}=\mathrm{Lie}(G)$ and $\mathfrak{h}=\mathrm{Lie}(T)$ again, we also recall from Section 3.4 the construction of the adjoint bundle $\widetilde{\mathfrak{g}}$, the adjoint quotient $\chi\colon\mathfrak{g}\to\mathfrak{h}/\!\!/W$ and $\widetilde{\chi}\colon\widetilde{\mathfrak{g}}\to\mathfrak{h}$. For a homogeneous (relative) transverse slice $\Sigma\subset\mathfrak{g}$ we set:

$$\widetilde{\Sigma}=\widetilde{\mathfrak{g}}\times_g\Sigma,\quad \chi_\Sigma=\chi|_\Sigma,\quad \widetilde{\chi}_\Sigma=\widetilde{\chi}|_{\widetilde{\Sigma}}$$

Proposition 4.5. *We have a commutative diagram with the following properties*

$$\begin{array}{ccccc}\widetilde{\Sigma} & \xrightarrow{q} & \Sigma\times_{\mathfrak{h}/\!/W}\mathfrak{h} & \longrightarrow & \Sigma\\ & \searrow^{\widetilde{\chi}_\Sigma} & \downarrow & & \downarrow{\scriptstyle\chi_\Sigma}\\ & & \mathfrak{h} & \xrightarrow{\psi} & \mathfrak{h}/\!/W\end{array}\tag{4.3}$$

(i) *χ_Σ is faithfully flat with geometrically normal surface fibers.*

(ii) *Let $Z=\chi_\Sigma^{-1}(0)=\Sigma\cap\mathcal{N}$, then for any geometric point $\bar{t}\to\mathrm{Spec}(\Lambda)$ we have that $Z_{\bar{t}}=\Sigma_{\bar{t}}\cap\mathcal{N}_{\bar{t}}$ has a unique rational double point singularity with associated root lattice $\Phi=\Phi(G,T)$.*

(iii) $\widetilde{\chi}_\Sigma$ *is smooth of relative dimension 2 and* q *is projective and induces the minimal resolution at every geometric fiber over* $\mathfrak{h}$.

Proof. The diagram (4.3) is obtained from restricting the Grothendieck alteration diagram (see Section 3.4) to $\Sigma \subset \mathfrak{g}$. For (i), note that $\alpha : G \times \Sigma \to \mathfrak{g}$ is smooth by Proposition 4.3 and since χ is G-invariant we have $\chi \circ \alpha = \chi_\Sigma \circ \mathrm{pr}_2$ where $\mathrm{pr}_2 : G \times \Sigma \to \Sigma$. By faithfully flat descent along pr_2 and Theorem 3.5 (ii) we get that χ_Σ is flat; faithfulness follows by noting the image of χ_Σ is open, invariant under the contraction action on $\mathfrak{h}/\!\!/W$ and contains the origin in every geometric fiber, so its complement is empty. As normality is preserved by smooth morphisms we get that the geometric fibers are normal of dimension $\dim_\Lambda(\Sigma) - r = 2$.

For (ii), we have that $x_{\bar{t}}$ is subregular nilpotent and $\Sigma_{\bar{t}}$ is a transverse slice ([Slo80b, §5.1]). Since $\mathrm{char}(k(\bar{t})) = 0$ or p very good for Φ, we have by [Slo80b, §6.4 Thm.] that $\Sigma_{\bar{t}} \cap \mathcal{N}_{\bar{t}}$ has a rational double point of type Φ at $x_{\bar{t}}$, and moreover by the contracting action μ defined in (4.2) it is the unique singularity of $Z_{\bar{t}}$.

For (iii), we have by Lemma 4.2 applied to the Grothendieck alteration $\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ that $\widetilde{\alpha} : G \times_\Lambda \widetilde{\Sigma} \to \widetilde{\mathfrak{g}}$ is smooth and moreover $\widetilde{\chi} \circ \widetilde{\alpha} = \widetilde{\chi}_\Sigma \circ \mathrm{pr}_2$ as before. By faithfully flat descent we conclude $\widetilde{\chi}_\Sigma$ is smooth. Projectivity of q follows from base-changing $\widetilde{\mathfrak{g}} \to \mathfrak{g} \times_{\mathfrak{h}/\!\!/W} \mathfrak{h}$ along $\Sigma \hookrightarrow \mathfrak{g}$. For the final assertion, let $\bar{t} \to \mathfrak{h}$ be a geometric point and form

$$q_{\bar{t}} : Y_{\bar{t}} := \widetilde{\chi}_\Sigma^{-1}(\bar{t}) \to X_{\bar{t}} := \chi_\Sigma^{-1}(\psi(\bar{t}))$$

$Y_{\bar{t}}$ is a smooth surface, and for $G_{\bar{t}} = G \times_\Lambda \bar{t}$, $\mathfrak{g}_{\bar{t}} = \mathfrak{g} \times_\Lambda \bar{t}$ and $\widetilde{\mathfrak{g}}_{\bar{t}} = \widetilde{\mathfrak{g}} \times_\Lambda \bar{t}$ we have $G_{\bar{t}}$-equivariant maps $\chi^{-1}(\psi(\bar{t})) \to \mathfrak{g}_{\bar{t}}$ and $\widetilde{\chi}^{-1}(\bar{t}) \to \widetilde{\mathfrak{g}}_{\bar{t}}$. Applying Lemma 4.2 we obtain smooth morphisms $G_{\bar{t}} \times X_{\bar{t}} \to \chi^{-1}(\psi(\bar{t}))$ and $G_{\bar{t}} \times Y_{\bar{t}} \to \widetilde{\chi}^{-1}(t)$. Thus $\mathrm{id}_{G_{\bar{t}}} \times q_{\bar{t}}$ is a smooth base-change of the resolution $\widetilde{\chi}^{-1}(\bar{t}) \to \chi^{-1}(\psi(\bar{t}))$ from Theorem 3.8 (iv), and so $q_{\bar{t}}$ is itself a resolution.

It remains to check minimality of $q_{\bar{t}}$. As both $\widetilde{\mathfrak{g}}/\Lambda$ and $\mathfrak{h}/\Lambda$ have trivial canonical bundle, we have $\omega_{\widetilde{\mathfrak{g}}/\mathfrak{h}} \simeq \mathcal{O}_{\widetilde{\mathfrak{g}}}$. By adjunction along the codimension c smooth subvariety $Y_{\bar{t}} \subset \widetilde{\mathfrak{g}}_{\bar{t}}$ we get

$$\omega_{Y_{\bar{t}}} \simeq \omega_{\widetilde{\mathfrak{g}}}|_{Y_{\bar{t}}} \otimes \det \mathcal{N}_{Y_{\bar{t}}/\widetilde{\mathfrak{g}}} \simeq \mathcal{O}_{Y_{\bar{t}}} \otimes \det \mathcal{O}_{Y_{\bar{t}}}^{\oplus c} \simeq \mathcal{O}_{Y_{\bar{t}}}$$

Thus for any exceptional rational curve C in $Y_{\bar{t}}$ we have $-2 = K_{Y_{\bar{t}}} \cdot C + C^2 = C^2$ and so there are no exceptional (-1)-curves, i.e. $q_{\bar{t}}$ is a minimal resolution. □

Proposition 4.5 says that $q : \widetilde{\Sigma} \to \Sigma \times_{h/\!\!/W} \mathfrak{h}$ is a simultaneous resolution of the rational double point surface family χ_Σ, after base-changing along finite morphism ψ.

In order to view the above as $\mathbb{G}_m$-equivariant deformations of singularities and their minimal resolutions, we need to equip the relevant spaces with compatible $\mathbb{G}_m$-actions as follows:

Recall the action μ on $\mathfrak{g}$ from (4.2), which preserves slice Σ. On $\widetilde{\mathfrak{g}}$ we define an action

$$\widetilde{\mu} : t \cdot [g, v] = [\lambda(t^{-1})g, t^2 v], \quad [g, v] \in (G \times^B \mathfrak{b})(\Lambda) \tag{4.4}$$

where $\lambda : \mathbb{G}_m \to G$ is the associated cocharacter defining Σ; this action preserves $\widetilde{\Sigma}$. Finally on $\mathfrak{h}$ we have the contracting action $t \cdot h = t^2 h$ for $h \in \mathfrak{h}(\Lambda)$ which commutes

with W and thus induces an action on $\mathfrak{h}/\!\!/W$; explicitly, one can choose homogeneous generators $p_1, \cdots, p_r$ of $\Lambda[\mathfrak{h}]^W$ by Theorem 3.5 (i) and the induced weights of the $\mathbb{G}_m$-action are $2\deg(p_i)$.

We next give a Lie-theoretic description of the exceptional divisors which appear in the minimal resolutions of Proposition 4.5 (iii), which will then be used in Theorem 4.7. For ease of notation, we work over the residue field k. Let $E \subset \widetilde{Z} \to Z$ be the reduced exceptional divisor of the minimal resolution of $Z = \chi_\Sigma^{-1}(0)$, which has a unique RDP by Proposition 4.5 (ii). Decompose $E = \cup_i E_i$ into its irreducible components.

Let $\{\alpha_i\}_{i \leq r}$ be the simple roots determined by B with associated minimal parabolics P_i such that $P_i/B \simeq \mathbb{P}^1_k$. Via the natural projection $\widetilde{\chi}^{-1}(0) = G \times^B \mathfrak{n} \to G/B$ we can identify $E_i \subset \widetilde{Z} \subset G \times^B \mathfrak{n}$ with its image P_i/B in G/B, up to translating by G (see [Slo80b, §6.3-6.4] for details).

On the other hand $\widetilde{Z} \subset \widetilde{\Sigma}$, and the normal bundle sequence ([EGA$_{\text{IV}}$, Prop. 19.1.5])

$$0 \to \mathcal{N}_{E_i/\widetilde{Z}} \to \mathcal{N}_{E_i/\widetilde{\Sigma}} \to \mathfrak{h} \otimes \mathcal{O}_{E_i} \to 0$$

yields a connecting homomorphism $\delta : \mathfrak{h} \to \mathrm{H}^1(E_i, \mathcal{N}_{E_i/\widetilde{Z}}) \simeq \mathrm{H}^1(\mathbb{P}^1_k, \mathcal{O}(-2)) \simeq k$.

Lemma 4.6. *There exist generators $e_i \in \mathrm{H}^1(E_i, \mathcal{N}_{E_i/\widetilde{Z}})$ such that $\delta(v) = d\alpha_i(v)e_i \ \forall v \in \mathfrak{h}$.*

Proof. Let us write $E = E_i, P_i = P$ and suppress the index i. We have identified E with $P/B \subset G/B$. Restricting the vector bundle sequence (3.3) from Section 3.4 to E yields

$$0 \to P \times^B \mathfrak{n} \to P \times^B \mathfrak{b} \to \mathfrak{h} \otimes \mathcal{O}_E \to 0 \tag{4.5}$$

Let U_P and L be the corresponding unipotent radical and Levi subgroup of P, $B_L = B \cap L$, $\mathfrak{b}_L = \mathrm{Lie}(B_L)$, $\mathfrak{n}_P = \mathrm{Lie}(U_P)$ and $\mathfrak{n}_L = \mathrm{Lie}(\mathcal{R}_u(B_L))$. We have identifications

$$E \simeq P/B \simeq L/B_L, \ P \times^B (\mathfrak{n}/\mathfrak{n}_P) \simeq L \times^{B_L} \mathfrak{n}_L, \ \ P \times^B (\mathfrak{b}/\mathfrak{n}_P) \simeq L \times^{B_L} \mathfrak{b}_L$$

via the quotient $P \to L$. Thus quotiening (4.5) by the trivial sub-bundle $P \times^B \mathfrak{n}_P \simeq E \times \mathfrak{n}_P$ yields exact sequence

$$0 \to L \times^{B_L} \mathfrak{n}_L \to L \times^{B_L} \mathfrak{b}_L \to \mathfrak{h} \otimes \mathcal{O}_E \to 0 \tag{4.6}$$

Moreover $\mathrm{H}^1(E, P \times^B \mathfrak{n}) \simeq \mathrm{H}^1(E, L \times^{B_L} \mathfrak{n}_L)$ so the connecting homomorphisms of (4.5) and (4.6) are identified.

Now the natural quotient $L \to L/Z(L) \simeq \mathrm{PGL}_2$ induces a differential map on the associated Cartan algebras $\mathfrak{h} = \mathfrak{b}_L/\mathfrak{n}_L \to k$ which is identified with $d\alpha$. The associated bundle sequence of $\mathfrak{pgl}_2$ is the Euler sequence

$$0 \to \mathcal{O}_{\mathbb{P}^1}(-2) \to \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2} \to \mathcal{O}_{\mathbb{P}^1} \to 0$$

and so the quotient $L \to \mathrm{PGL}_2$ induces a map of associated bundle sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & L\times^{B_L}\mathfrak{n}_L & \longrightarrow & L\times^{B_L}\mathfrak{b}_L & \longrightarrow & \mathfrak{h}\otimes\mathcal{O}_E & \longrightarrow & 0 \\ & & \downarrow\simeq & & \downarrow & & \downarrow d\alpha & & \\ 0 & \longrightarrow & \mathcal{O}(-2) & \longrightarrow & \mathcal{O}(-1)^{\oplus 2} & \longrightarrow & \mathcal{O} & \longrightarrow & 0 \end{array}$$

where on the left we identify $L\times^{B_L}\mathfrak{n}_L$ with the cotangent bundle of $E\simeq\mathbb{P}^1$. Since the connecting homomorphism of the bottom row is isomorphism $k\xrightarrow{\sim}\mathrm{H}^1(E,\mathcal{O}_E(-2))$, by commutativity of the right side of the diagram we get that the connecting homomorphism $\partial:\mathfrak{h}\to\mathrm{H}^1(E,\mathcal{O}(-2))$ in the top row is $d\alpha$; this is also the connecting homomorphism in (4.5). It remains to show that this connecting homomorphism coincides with $\delta:\mathfrak{h}\to\mathrm{H}^1(E,\mathcal{N}_{E/\widetilde{Z}})$.

Writing $\Sigma=x+\mathfrak{a}$ for the relative transverse slice, we have a normal bundle sequence

$$0\to\mathcal{N}_{E/\widetilde{Z}}\to\mathcal{N}_{E/G\times^B\mathfrak{n}}\to\mathfrak{g}/\mathfrak{a}\otimes\mathcal{O}_E\to 0$$

which splits since $\mathfrak{g}=[\mathfrak{g},x]\oplus\mathfrak{a}$ and hence yields an isomorphism

$$\mathrm{H}^1(E,\mathcal{N}_{E/G\times^B\mathfrak{n}})\simeq\mathrm{H}^1(E,\mathcal{N}_{E/\widetilde{Z}})$$

Similarly via the normal bundle sequence associated to vector bundle $G\times^B\mathfrak{n}\to G/B$ we have

$$0\to P\times^B\mathfrak{n}\to\mathcal{N}_{E/G\times^B\mathfrak{n}}\to\mathcal{N}_{E/(G/B)}\to 0$$

where $E\simeq P/B$; since $\mathcal{N}_{E/(G/B)}$ is trivial, we get an isomorphism $\mathrm{H}^1(E,P\times^B\mathfrak{n})\simeq\mathrm{H}^1(E,\mathcal{N}_{E/G\times^B\mathfrak{n}})$. Altogether, this identifies the connecting homomorphisms ∂ and δ via $\delta(v)=\partial(v)=d\alpha(v)e$ after setting $\mathrm{H}^1(E,P\times^B\mathfrak{n})\simeq\mathrm{H}^1(E,\mathcal{O}(-2))\simeq k\langle e\rangle$. □

Theorem 4.7. *Assume p is very good for $\Phi(G,T)$ and consider a homogeneous transverse slice $\Sigma\subset\mathfrak{g}$ at subregular nilpotent x with maps $\chi_\Sigma,\widetilde{\chi}_\Sigma$ as above. Let $\bar{t}=\mathrm{Spec}(k)\to\Lambda$ be the closed point of Λ and set $Z=\chi_\Sigma^{-1}(0)$, $R=\widehat{\mathcal{O}}_{Z_k,x_k}$, $A=\widehat{\mathcal{O}}_{h/\!\!/W,0}$ and $B=\widehat{\mathcal{O}}_{\mathfrak{h},0}$. Suppose X is the minimal resolution of $\mathrm{Spec}(R)$ with exceptional divisor E and $\mathcal{X}:=\widehat{X_E}$. Denote the completions of χ_Σ and $\widetilde{\chi}_\Sigma$ along x_k and E respectively by $\widehat{\chi}_\Sigma$ and $\widehat{\widetilde{\chi}}_\Sigma$.*

(i) *$\widehat{\chi}_\Sigma$ is an equivariantly miniversal deformation of R over $\mathrm{Spf}(A)$.*

(ii) *$\widehat{\widetilde{\chi}}_\Sigma$ is a miniversal deformation of $\mathcal{X}$ over $\mathrm{Spf}(B)$.*

Proof. We have $A\simeq\Lambda[\![p_1,\cdots p_r]\!],B\simeq\Lambda[\![h_1,\cdots,h_r]\!]$ for homogeneous coordinates p_i,h_i, thus by Proposition 2.17 for $\widehat{\chi}_\Sigma$ and Proposition 2.25 for $\widehat{\widetilde{\chi}}_\Sigma$ it suffices to show that their respective Kodaira–Spencer maps are isomorphisms.

Consider first $\mathrm{KS}_{\chi_\Sigma}:T_\Lambda A\to\mathrm{Def}_R(k[\varepsilon]/\varepsilon^2)\simeq T^1(R)$, and write $C=\widehat{\mathcal{O}}_{\Sigma,x_k}$ so that $\widehat{\chi}_\Sigma:\mathrm{Spf}(C)\to\mathrm{Spf}(A)$. Then

$$\mathrm{KS}_{\chi_\Sigma}(v)=[C\widehat{\otimes}_{A,v}k[\varepsilon]/\varepsilon^2]=[(C/\mathfrak{m}_\Lambda C)\otimes_{A/\mathfrak{m}_\Lambda A,v}k[\varepsilon]/\varepsilon^2]$$

and consequently it suffices to show that the Kodaira–Spencer map of the characteristic p deformation $\mathrm{Spf}(C/\mathfrak{m}_\Lambda C)\to\mathrm{Spf}(A/\mathfrak{m}_\Lambda A)\simeq\mathrm{Spf}(k[\![p_1,\cdots,p_r]\!])$ is an isomorphism.

We therefore work over k from now on. Using Proposition 2.23, we obtain a $\mathbb{G}_m$-equivariantly miniversal deformation as follows: write $R = k[\![x,y,z]\!]/f(x,y,z)$ for f as in Theorem 2.2 (ii), choose homogeneous generators $\{g_i\}$ of $T^1(R)$ and then complete the family $F : V(f - \sum t_i g_i) \to \mathrm{Spec}(k[t_1, \cdots, t_r])$ for the desired equivariantly miniversal deformation. By [Slo80b, §7.4, Prop. 2, §8.1, Lem. 2] the weights of the t_i for the $\mathbb{G}_m$-action may be chosen to be $2\deg(p_i)$.

Deformation $\widehat{F}$ induces a classifying map $\widehat{\phi} : \mathrm{Spf}(k[\![p_1, \cdots, p_r]\!]) \to \mathrm{Spf}(k[\![t_1, \cdots, t_r]\!])$ with the same $\mathbb{G}_m$-weights on parameters p_i, t_i, and moreover since these weights are positive, the map is algebraic i.e. it is induced by a polynomial map

$$\phi : \mathrm{Spec}(k[p_1, \cdots, p_r]) \to \mathrm{Spec}(k[t_1, \cdots, t_r])$$

so that $\chi_\Sigma \otimes k$ is the pullback of the algebraic family F.

It now suffices to show $\phi^{-1}(0)_{\mathrm{red}} = 0$: if $h \in (\mathfrak{h}/\!\!/W)_k$ has $\phi(h) = 0$ then $\chi_\Sigma^{-1}(h)$ has a rational double point with associated root lattice Φ, and lifting h to $y \in \mathfrak{h}_k$ we have that the singularity of $\chi_\Sigma^{-1}(\psi(y))$ has as root lattice a connected component of the $\mathbb{Z}$-lattice $\Phi_y = \langle \alpha \in \Phi \mid d\alpha(y) = 0 \rangle$ ([Slo80b, §6.5, Lem. 1]). So $\Phi_y = \Phi$, but in very good characteristic the simple root differentials $d\alpha_i$ form a basis of $\mathfrak{h}_k^\vee$ so $y = h = 0$. We conclude by [Slo80b, §8.1, Lem 3] that ϕ is an isomorphism and so $\mathrm{KS}_{\chi_\Sigma} \otimes_\Lambda k = \mathrm{KS}_F \circ d\phi_0$ is an isomorphism, i.e. $\mathrm{KS}_{\chi_\Sigma}$ is an isomorphism and we deduce (i).

For (ii), we need to show that the mod p Kodaira–Spencer map is an isomorphism, as in (i). Identifying $\mathrm{Def}_{\mathcal{X}}(k[\varepsilon]/\varepsilon^2) \simeq \mathrm{H}^1(\mathcal{X}, T_{\mathcal{X}})$ as in Section 2.4 and $T_\Lambda(B/\mathfrak{m}_\Lambda B) \simeq \mathfrak{h}_k \simeq k^r$, we have by Proposition 2.25 a composition

$$k^r \xrightarrow{\mathrm{KS}} \mathrm{H}^1(\mathcal{X}, T_{\mathcal{X}}) \xrightarrow{\rho} \bigoplus_{i=1}^{r} H^1(E_i, \mathcal{N}_{E_i/X})$$

where ρ is an isomorphism and E_i are the smooth exceptional curves in the minimal resolution $X \to \mathrm{Spec}(R)$. By construction of the Kodaira–Spencer map, setting ρ_i for the i-th component of ρ we have that $\rho_i \circ \mathrm{KS}$ is identified with connecting homomorphism $\delta : \mathfrak{h}_k \to \mathrm{H}^1(E_i, \mathcal{N}_{E_i/X})$ of the normal bundle sequence associated to $E_i \subset X \subset \widetilde{\Sigma}$; see the paragraph before Lemma 4.6.

Let $\{e_i\}$ generate the 1-dimensional spaces $\mathrm{H}^1(E_i, \mathcal{N}_{E_i/X})$ and let $\alpha_i \in \Phi$ be the simple root corresponding to E_i. By Lemma 4.6 we have $\rho \circ \mathrm{KS}(v) = (d\alpha_i(v)e_i)_i$, so $\rho \circ \mathrm{KS}$ is a linear map $v \mapsto (d\alpha_i(v))_i$, and in the usual simple coroot basis it is given by the Cartan matrix, whose determinant is invertible in very good characteristic. We conclude $\mathrm{KS}_{\widehat{\chi}_\Sigma}$ is an isomorphism and so we get (ii). □

As a result, the completed map $\widehat{q}$ is a simultaneous resolution of $\widehat{\chi}_\Sigma$ after pullback along $\mathrm{Spf}(B) \to \mathrm{Spf}(A)$.

5. The monodromy Weyl action

The main goal of this section is the proof of Theorem 1.1 (i.e. Theorem 5.16). To this end, Sections 5.1-5.3 cover prerequisites like nearby cycles, Springer fiber cohomology and relative perversity, which are then used to define and compare the monodromy and Springer actions in Section 5.4.

5.1 Nearby and vanishing cycles. Throughout we fix a mixed characteristic complete[6] discrete valuation ring Λ with algebraically closed point $s = \mathrm{Spec}(k)$ and generic point $\eta = \mathrm{Spec}(K)$, and we write $D^b(X, \mathbb{Q}_\ell), D^b_c(X, \mathbb{Q}_\ell)$ for the bounded derived category of $\mathbb{Q}_\ell$ sheaves (respectively constructible $\mathbb{Q}_\ell$-sheaves) on noetherian finite type Λ-schemes X ($\ell \neq p$).

Definition 5.1. Let X be a finite-type Λ-scheme with special fiber $i : X_s \to X$, generic fiber $j : X_\eta \to X$ and geometric generic fiber $\bar{j} : X_{\overline{\eta}} \to X$.

(i) The *nearby cycles* functor is $\mathrm{R}\Psi_X : \mathrm{D}^b(X_\eta, \mathbb{Q}_\ell) \to \mathrm{D}^b(X_s, \mathbb{Q}_\ell)$, $\mathcal{F} \longmapsto i^* \mathrm{R}\bar{j}_*(\mathcal{F}_{\overline{\eta}})$, where $\mathcal{F}_{\overline{\eta}}$ is the pullback of $\mathcal{F}$ to $X_{\overline{\eta}}$. $\mathrm{R}\Psi_X(\mathcal{F})$ is naturally equipped with an action of inertia $I_K = \mathrm{Gal}_K = \mathrm{Gal}(\overline{\eta}/\eta)$ ([SGA 7_{II}], Exposé XIII, §1.3).

(ii) For $\mathcal{F} \in \mathrm{D}^b(X, \mathbb{Q}_\ell)$, adjunction $\mathcal{F} \to \mathrm{R}\bar{j}_*(\mathcal{F}_{\overline{\eta}})$ gives an exact I_K-equivariant triangle

$$i^*\mathcal{F} \xrightarrow{\phi} \mathrm{R}\Psi_X(\mathcal{F}|_{X_\eta}) \longrightarrow \mathrm{R}\Phi_X(\mathcal{F}) \xrightarrow{[1]} \tag{5.1}$$

defining the *vanishing cycles functor* $\mathrm{R}\Phi : \mathrm{D}^b(X, \mathbb{Q}_\ell) \to \mathrm{D}^b(X_s, \mathbb{Q}_\ell)$ as the cone of map ϕ. For its functoriality and equivariance see [SGA 7_{II}, Exposé XIII, 1.4.2, 2.1.1-2.1.2].

We record two relevant properties of nearby cycles. Given a map $f : X \to Y$ with induced maps f_η, f_s on the geometric generic, resp. special fibers, we have:

(i) For f proper, $\mathcal{F} \in \mathrm{D}^b_c(X_\eta, \mathbb{Q}_\ell)$, a natural isomorphism $\mathrm{R}\Psi_Y(\mathrm{R}f_{\eta*}\mathcal{F}) \xrightarrow{\sim} \mathrm{R}f_{s*}\mathrm{R}\Psi_X(\mathcal{F})$

(ii) For f smooth, $\mathcal{G} \in D^b_c(Y_\eta, \mathbb{Q}_\ell)$, a natural isomorphism $f_s^*\mathrm{R}\Psi_Y(\mathcal{G}) \xrightarrow{\sim} \mathrm{R}\Psi_X(f_\eta^*\mathcal{G})$

In particular, when $f : X \to \mathrm{Spec}(\Lambda)$ is smooth we have $\mathrm{R}\Psi_X\mathbb{Q}_\ell \simeq f_s^*\mathrm{R}\Psi_\Lambda\mathbb{Q}_\ell \simeq \mathbb{Q}_\ell$ on X_s so $\mathrm{R}^{\bullet \geq 1}\Psi_X\mathbb{Q}_\ell = \mathrm{R}\Phi_X\mathbb{Q}_\ell = 0$; more generally, $\mathrm{R}\Phi_X\mathbb{Q}_\ell$ and $\mathrm{R}^{\bullet\geq 1}\Psi_X\mathbb{Q}_\ell$ vanish on the smooth locus of X/Λ (see [SGA 7_{II}, Exposé XIII, 2.1.5, 2.1.7.1-2]).

Given $\overline{x} \to X_s$ and setting $X_{(\overline{x})} := \mathrm{Spec}\left(\mathcal{O}^{\mathrm{sh}}_{X,\overline{x}}\right)$, the nearby cycle stalks are

$$(\mathrm{R}\Psi_X\mathcal{F})_{\overline{x}} \simeq \mathrm{R}\Gamma(X_{(\overline{x})} \times_\Lambda \overline{\eta}, \mathcal{F}_\eta)$$

by [SGA 7_{II}, Exposé XIII, Prop. 2.1.4]. We call $X_{(\overline{x})}$ the ℓ-adic *Milnor ball* around x and $X_{(\overline{x})} \times_\Lambda \overline{\eta}$ the ℓ-adic *Milnor fiber* at x, which alludes to the topological fact that complex-analytic nearby cycle stalks compute the singular cohomology of Milnor fibers of isolated singularities.

Example 5.2. Let X be a flat proper surface over Λ, smooth outside an isolated rational double point on the special fiber $x \to X_s$, then we have ([SGA 7_{II}], Exposé XVI)

$$(R^0\Psi_X\mathbb{Q}_\ell)_x \simeq \mathbb{Q}_\ell, \quad (\mathrm{R}^2\Psi_X\mathbb{Q}_\ell)_x \simeq \mathbb{Q}_\ell^r, \quad (\mathrm{R}^i\Psi_X\mathbb{Q}_\ell)_x = 0 \text{ for } i \neq 0, 2$$

where r is the rank of the root lattice associated to RDP $\overline{x}$ (see Theorem 2.2).

[6] For this subsection, Λ strictly henselian suffices.

Proposition 5.3. *Assume Λ is complete and let X, Y be finite-type Λ-schemes with closed points $\overline{x} \to X_s, \overline{y} \to Y_s$ on their special fibers. A chosen isomorphism $\widehat{\mathcal{O}}_{X,x} \simeq \widehat{\mathcal{O}}_{Y,y}$ of complete local Λ-algebras induces an I_K-equivariant isomorphism*

$$(R\Psi_X \mathbb{Q}_\ell)_x \xrightarrow{\sim} (R\Psi_Y \mathbb{Q}_\ell)_y$$

Proof. Let $(U, u) \to (X, x)$ be a pointed étale neighborhood, then by invariance of strict henselizations under étale maps we have $U_{(u)} \simeq X_{(x)}$ ([Stacks, Tag 094Z]). Thus we have an isomorphism

$$(\mathrm{R}\Psi_X \mathbb{Q}_\ell)_x \simeq \mathrm{R}\Gamma(X_{(x)} \times_\Lambda \overline{\eta}, \mathbb{Q}_\ell) \simeq \mathrm{R}\Gamma(U_{(u)} \times_\Lambda \overline{\eta}, \mathbb{Q}_\ell) \simeq (\mathrm{R}\Psi_U \mathbb{Q}_\ell)_u$$

compatible with the I_K-action on the stalks since the strict henselizations are defined over $\mathrm{Spec}(\Lambda)$. As complete noetherian Λ-algebras are G-rings, by Artin approximation ([Stacks, Tag 0CAV]) we can find a common pointed étale neighborhood

$$(X, x) \leftarrow (U, u) \to (Y, y)$$

and consequently we get an I_K-equivariant isomorphism $(\mathrm{R}\Psi_X \mathbb{Q}_\ell)_x \simeq (\mathrm{R}\Psi_Y \mathbb{Q}_\ell)_y$. By [Ber96, Thm. 3.1] this isomorphism is induced by $\widehat{\mathcal{O}}_{X,x} \simeq \widehat{\mathcal{O}}_{Y,y}$. □

We remark that Proposition 5.3 also applies to stalks of vanishing cycles as the isomorphism above is compatible with specialization $\mathbb{Q}_\ell \to \mathrm{R}\Psi_X \mathbb{Q}_\ell$, and moreover it suffices to have a common étale neighborhood of (X, x) and (Y, y) (cf. Corollary 2.21). The underlying point of the functoriality of the nearby cycles stalk isomorphism is that there is a canonical isomorphism

$$(\mathrm{R}\Psi_X \mathbb{Q}_\ell)_x \simeq \mathrm{R}\Gamma(\mathcal{M}_x, \mathbb{Q}_\ell)$$

where $\mathcal{M}_x := (\mathrm{Spf}(\widehat{\mathcal{O}}_{X,x}))_\eta \widehat{\times}_K \widehat{\overline{K}}$ is the analytic Milnor fiber (in the sense of [Ber96, §1]).

5.2 ℓ-adic Springer correspondences. As in Section 3.1 we fix a split, simply connected, simple group G/Λ and a split maximal torus $T \subset G$, and set $\mathfrak{g} = \mathrm{Lie}(G), \mathfrak{h} = \mathrm{Lie}(T)$. In particular we retain Assumption 3.1 regarding the residue characteristic. We also recall from Section 3.4 the construction of the adjoint bundle $\widetilde{\mathfrak{g}}$ and the Grothendieck alteration $\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$.

For a geometric point $\overline{t} \to \Lambda$ consider the pullbacks $\mathfrak{g}_{\overline{t}} = \mathfrak{g} \times_\Lambda \overline{t}, \widetilde{\mathfrak{g}}_{\overline{t}} = \widetilde{\mathfrak{g}} \times_\Lambda \overline{t}$ and $\pi_{\overline{t}} : \widetilde{\mathfrak{g}}_{\overline{t}} \to \mathfrak{g}_{\overline{t}}$ the classical Grothendieck alteration; $\mathfrak{g}_{\overline{t}}$ is a Lie algebra over a field of zero or very good characteristic, as per our assumptions. We omit the subscript $(\cdot)_{\overline{t}}$ in this subsection for ease of notation, and assume $\mathfrak{g}$ is a Lie algebra over $k = \overline{k}$.

Definition 5.4. For $x \in \mathcal{N}$, the *Springer fiber* over x is $\mathcal{B}_x = \pi^{-1}(x)_{\mathrm{red}} \subset \widetilde{\mathfrak{g}}$.

By the incidence variety description of $\widetilde{\mathfrak{g}}$ (see Remark 3.7) we have that $\mathcal{B}_x$ is the projective variety of Borels B' so that $x \in \mathrm{Lie}(B')$; in particular $\mathcal{B}_0 = G/B$.

Let W be the Weyl group associated to the root lattice $\Phi = \Phi(G, T)$. There is a natural construction of Springer ([Spr76], see also [Sho88, §6]) which gives a correspondence between irreducible $\overline{\mathbb{Q}_\ell}$-representations of W and nilpotent orbits in $\mathcal{N}$ equipped

with an equivariant irreducible local system; under this correspondence, the cohomology of Springer fibers $\mathrm{H}^i(\mathcal{B}_x, \mathbb{Q}_\ell)$ is endowed with a W-action. The following sheaf-theoretic construction of this action is due to Borho–MacPherson ([BM83, §2.5-2.6], see also [Sho88, §4-§6]):

Over the regular semisimple locus $\mathfrak{g}^{\mathrm{rs}}$ (Section 3.2) the Grothendieck alteration is a W-torsor $\pi^{\mathrm{rs}} : \widetilde{\mathfrak{g}}^{\mathrm{rs}} \to \mathfrak{g}^{\mathrm{rs}}$. More explicitly, isomorphism $\widetilde{\mathfrak{g}}^{\mathrm{rs}} \simeq G/T \times_\Lambda \mathfrak{h}^{\mathrm{rs}}$ equips the left hand side with a W-action by picking a representative $\widetilde{w} \in N_G(T)$ of w and setting $a_w(gT, h) = (g\widetilde{w}^{-1}T, \mathrm{Ad}(\widetilde{w})h)$ (Theorem 3.8 (ii) and (iii)). We remark that this formula does not depend on the choice of $\widetilde{w}$. Since $\mathrm{Ad}(g\widetilde{w}^{-1})\mathrm{Ad}(\widetilde{w})h = \mathrm{Ad}(g)h$, the W-torsor structure on π^{rs} is clear by $\pi^{\mathrm{rs}} \circ a_w = \pi^{\mathrm{rs}}$. Setting $\mathcal{L} := \pi^{\mathrm{rs}}_*\mathbb{Q}_\ell$ we therefore have a natural map $\mathbb{Q}_\ell[W] \to \mathrm{End}(\mathcal{L})$ via $w \cdot v(x) = v(a_w^{-1}x)$ on functions v on its (finite) stalks.

Now $\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ is a small morphism (in the sense of [Sho88, Rem. 4.4]) and so setting $d = \dim \mathfrak{g}$ we have (Prop. 4.2 *loc. cit.*)

$$\mathrm{R}\pi_*\mathbb{Q}_\ell[d] \simeq j_{!*}(\pi^{\mathrm{rs}}_*\mathbb{Q}_\ell[d]) =: \mathrm{IC}_{\mathfrak{g}}(\mathcal{L}) \tag{5.2}$$

where $j : \mathfrak{g}^{\mathrm{rs}} \hookrightarrow \mathfrak{g}$. In other words $\mathrm{R}\pi_*\mathbb{Q}_\ell[d]$ is an IC (perverse) sheaf on $\mathfrak{g}$, and full-faithfullness of the intermediate extension $j_{!*}$ extends every deck transformation of W uniquely to $\mathrm{R}\pi_*\mathbb{Q}_\ell[d]$. Proper base-change along $i_x : x \to \mathcal{N} \subset \mathfrak{g}$ yields

$$(\mathrm{R}\pi_*\mathbb{Q}_\ell[d])_x[-d] \simeq \mathrm{R}\Gamma(\mathcal{B}_x, \mathbb{Q}_\ell)$$

and the right-hand side is naturally equipped with the *Springer* W-action; the convention here is that when x is regular (whence $\mathcal{B}_x$ is discrete) the associated W-representation $\mathrm{H}^0(\mathcal{B}_x, \mathbb{Q}_\ell)$ is the trivial one.

We specialize to studying W-representations for $\mathcal{B}_x$ when x is subregular: denote by $\alpha_1, \cdots, \alpha_r$ the simple roots of $\mathfrak{h}$ (here $r = \mathrm{rk}(\mathfrak{g})$). W has a natural action on the character lattice $X^*(T)$ and its $\mathbb{Q}_\ell$-linear extension $V_\ell := X^*(T) \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$ via the reflecton representation: explicitly, the simple reflections $s_{\alpha_i} \in W$ yield $s_{\alpha_i}(\lambda) = \lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i$ for $\lambda \in X^*(T)$.

Viewing λ as a character $B \to T \to \mathbb{G}_m$, we set $k_\lambda := k$ equipped with a B-action so that $b \in B$ acts as multiplication by $\lambda(b)^{-1}$. We form the associated (line) bundle

$$\mathcal{L}_\lambda := G \times^B k_\lambda \tag{5.3}$$

It is a line bundle over G/B. If P_i/B is the minimal parabolic associated to α_i, we have $\mathcal{L}_\lambda|_{P_i/B} \simeq \mathcal{O}_{\mathbb{P}^1}(\langle \lambda, \alpha_i^\vee \rangle)$ (cf. [BK05, §2.1]). Moreover we have a natural map

$$X^*(T) \to \mathrm{Pic}(\mathcal{B}_x), \quad \lambda \mapsto [\mathcal{L}_\lambda|_{\mathcal{B}_x}]$$

and thus by $\mathbb{Q}_\ell$-linear extension an ℓ-adic Chern class map $c : V_\ell \to \mathrm{H}^2(\mathcal{B}_x, \mathbb{Q}_\ell(1))$ sending $\lambda \otimes 1 \mapsto c_1(\mathcal{L}_\lambda|_{\mathcal{B}_x})$.

Proposition 5.5. *Let x be subregular, then the Chern class map $c : V_\ell \to \mathrm{H}^2(\mathcal{B}_x, \mathbb{Q}_\ell(1))$ is a W-equivariant isomorphism, where the W-action on the source is the reflection representation.*

Proof. The Chern class map $\widetilde{c} : V_\ell \to \mathrm{H}^2(G/B, \mathbb{Q}_\ell(1))$, $\lambda \mapsto [\mathcal{L}_\lambda]$ is W-equivariant with respect to the 'classical' W-action on the target as defined in [Sho88, §5.1], and this action is equivalent to the Springer W-action by [Sho88, Prop. 5.4]. Moreover the proof of *loc. cit* shows that $\mathrm{H}^2(G/B, \mathbb{Q}_\ell) \to \mathrm{H}^2(\mathcal{B}_x, \mathbb{Q}_\ell)$ is W-equivariant; the composite is c after a Tate twist, and W-equivariance of this map follows.

Now $\mathcal{B}_x = (\pi^{-1}(x))_{\mathrm{red}} = \cup_{i \le r} E_i$ is a union of projective lines E_i: we can choose a suitable transverse slice $\Sigma = x + \mathfrak{a}$ at x (Section 4.1) so that $x \in \Sigma \cap \mathcal{N}$ is a rational double point of Dynkin type Φ, and $\pi^{-1}(x)$ is the exceptional divisor of its minimal resolution (Proposition 4.5 (ii) and (iii)). Up to translating by G we can view E_i as P_i/B (see Section 4.2). Then restriction to the irreducible components yields

$$\mathrm{H}^2(\mathcal{B}_x, \mathbb{Q}_\ell(1)) \xrightarrow{\sim} \bigoplus_{i=1}^{r} \mathrm{H}^2(E_i, \mathbb{Q}_\ell(1)) \xrightarrow{\sim} \mathbb{Q}_\ell^r$$

where the maps on the right are trace maps. Since $\mathcal{L}_\lambda|_{P_i/B} \simeq \mathcal{O}_{\mathbb{P}^1}(\langle \lambda, \alpha_i^\vee \rangle)$, map c sends $\lambda \in V_\ell \mapsto \oplus_i c_1\left(\mathcal{L}_\lambda|_{P_i/B}\right) = (\langle \lambda, \alpha_i^\vee \rangle)_{i \le r}$ and by picking λ generating $X^*(T)$ we see that c maps linear basis to linear basis, hence is an isomorphism. □

5.3 Relative perversity of the Grothendieck–Springer sheaf. We retain the setting of Sections 5.1, 5.2 from here and onwards, and consider constructible ℓ-adic sheaves on $\widetilde{\mathfrak{g}}$ and $\mathfrak{g}$ viewed as Λ-schemes of relative dimension d. To interpolate between points $\bar{t} \to \Lambda$ of different characteristic, it will be useful to consider perverse sheaves in families over Λ following Hansen–Scholze ([HS23]).

Definition 5.6. Let X be a separated finite-type Λ-scheme. A sheaf $\mathcal{F} \in D_c^b(X, \mathbb{Q}_\ell)$ is

(i) Λ-*perverse* if for every geometric point $\bar{t} \to \mathrm{Spec}(\Lambda)$ we have $\mathcal{F}|_{X_{\bar{t}}} \in \mathbf{Perv}(X_{\bar{t}}, \mathbb{Q}_\ell)$.

(ii) Λ-*universally locally acyclic* (or Λ-ULA) if for every specialization[7] $\bar{t} \rightsquigarrow \bar{s}$ in $\mathrm{Spec}(\Lambda)$ we have

$$\mathcal{F}_{\overline{x}} \xrightarrow{\sim} \mathrm{R}\Gamma\left(X_{(x)} \times_{\mathrm{Spec}(\Lambda)_{(s)}} \bar{t}, \mathcal{F}\right) \tag{5.4}$$

where $\overline{x}$ lives over $\overline{s}$, and moreover (5.4) holds after any base-change $T \to \mathrm{Spec}(\Lambda)$.

The objects in (i) form the heart $\mathbf{Perv}(X/\Lambda)$ of the relative perverse t-structure on $D_c^b(X, \mathbb{Q}_\ell)$; see [HS23, Thm. 1.1] for the definition of this category and this t-structure, which interpolates between the standard t-structure (when $X = \Lambda$) and the middle perverse t-structure (when $\Lambda = \mathrm{Spec}(\overline{k})$). In particular $\mathbf{Perv}(X_{\bar{t}}, \mathbb{Q}_\ell)$ for schemes over field $k(\bar{t})$ is always defined in terms of the middle perverse t-structure.

The objects in (ii) are also well-defined and those satisfying both (i) and (ii) form an abelian subcategory $\mathbf{Perv}^{\mathrm{ULA}}(X/\Lambda) \subset \mathbf{Perv}(X/\Lambda)$ ([HS23, Thm. 1.6 (iii), Thm. 1.9]).

Proposition 5.7. *Let $\pi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ be the Grothendieck alteration, $j : \mathfrak{g}^{rs} \hookrightarrow \mathfrak{g}$ the inclusion of the regular semisimple locus and $\mathcal{G} := \mathrm{R}\pi_* \mathbb{Q}_\ell[d]$ the* Grothendieck–Springer sheaf.

[7] By specialization we mean a choice of map $\bar{t} \to \mathrm{Spec}\left(\mathcal{O}^{\mathrm{sh}}_{\mathrm{Spec}(\Lambda), \bar{s}}\right)$, see [Stacks, Tag 0GJ2].

(i) $\mathcal{G}$ *is* Λ*-perverse and* Λ*-universally locally acyclic.*

(ii) *If* $\mathcal{L} = \pi_*^{rs}\mathbb{Q}_\ell$ *then* $\mathcal{G}$ *is a relative IC extension of* $\mathcal{L}[d]$ *in the sense that for every* $\bar{t} \to \mathrm{Spec}(\Lambda)$ *we have* $\mathcal{G}|_{\mathfrak{g}_{\bar{t}}} = (j_{\bar{t}})_{!*}(\mathcal{L}|_{\mathfrak{g}^{rs}_{\bar{t}}}[d])$.

(iii) *We have an isomorphism* $\mathrm{End}_{D^b_c(\mathfrak{g})}(\mathcal{G}) \xrightarrow{\sim} \mathrm{End}_{D^b_c(\mathfrak{g}^{rs})}(j^*\mathcal{G})$.

Proof. Since $\widetilde{\mathfrak{g}}/\Lambda$ is smooth, constant sheaf $\mathbb{Q}_\ell \in D^b_c(\widetilde{\mathfrak{g}})$ is Λ-ULA, and so by properness of π we have that $\mathcal{G}$ is Λ-ULA as well ([HS23, Prop. 3.3]).

For a geometric point $\bar{t} \to \mathrm{Spec}(\Lambda)$, by proper base-change we have $\mathcal{G}|_{\mathfrak{g}_{\bar{t}}} \simeq \mathrm{R}\pi_{\bar{t}*}\mathbb{Q}_\ell[d]$, the usual Grothendieck–Springer sheaf associated with Grothendieck alterations over fields (see Section 5.2). By the construction in Section 5.2 we have

$$\mathrm{R}\pi_{\bar{t}*}\mathbb{Q}_\ell[d] \simeq (j_{\bar{t}})_{!*}(\mathcal{L}|\mathfrak{g}^{\mathrm{rs}}_{\bar{t}}[d]), \quad j_{\bar{t}} : \mathfrak{g}^{\mathrm{rs}}_{\bar{t}} \hookrightarrow \mathfrak{g}_{\bar{t}}$$

This automatically makes $\mathcal{G}|_{\mathfrak{g}_{\bar{t}}}$ perverse (in fact an IC sheaf) so we get Λ-perversity of $\mathcal{G}$ and (ii).

For (iii), $\mathrm{Spec}(\Lambda)$ is integral and geometrically unibranch so by [HS23, Thm.1.10 (ii)] the restriction functor $\mathbf{Perv}^{\mathrm{ULA}}(\mathfrak{g}/\Lambda) \to \mathbf{Perv}(\mathfrak{g}_\eta)$ is exact and fully-faithful, where $\eta = \mathrm{Spec}(\mathrm{Frac}(\Lambda))$ is the generic point. Applying this fact to $\mathfrak{g}$ and $\mathfrak{g}^{\mathrm{rs}}$ we get

$$\mathrm{End}_{\mathfrak{g}}(\mathcal{G}) \simeq \mathrm{End}_{\mathfrak{g}_\eta}(\mathcal{G}|_{\mathfrak{g}_\eta}) \overset{(*)}{\simeq} \mathrm{End}_{\mathfrak{g}^{\mathrm{rs}}_\eta}(\mathcal{L}|_{\mathfrak{g}^{\mathrm{rs}}_\eta}) \simeq \mathrm{End}_{\mathfrak{g}^{\mathrm{rs}}}(\mathcal{L})$$

where $(*)$ stems from the fully-faithfulness of the usual intermediate extension functor $(j_\eta)_{!*} : \mathbf{Perv}(\mathfrak{g}^{\mathrm{rs}}_\eta) \to \mathbf{Perv}(\mathfrak{g}_\eta)$ (see Section 5.2); note $\mathcal{L}[d]$ is trivially Λ-ULA as a local system on the Λ-smooth scheme $\mathfrak{g}^{\mathrm{rs}}$. The composite is restriction along $j : \mathfrak{g}^{\mathrm{rs}} \hookrightarrow \mathfrak{g}$ and $\mathcal{L}[d] = j^*\mathcal{G}$, whence we get (iii). □

Now let $e \in \mathcal{N}(\Lambda)$ be fiberwise subregular. As we are in very good residue characteristic, we may construct a relative homogeneous transverse slice Σ at e (Section 4.1) of relative dimension $r+2$. Following the notation of Section 4.2 we set $i_\Sigma : \Sigma \hookrightarrow \mathfrak{g}$, $j_\Sigma : \Sigma^{\mathrm{rs}} = \Sigma \cap \mathfrak{g}^{\mathrm{rs}} \hookrightarrow \Sigma$ and $\pi_\Sigma : \widetilde{\Sigma} \xrightarrow{q} \Sigma \times_{\mathfrak{h}/\!/W} \mathfrak{h} \to \Sigma$ the slice-restricted Grothendieck alteration, with π^{rs}_Σ its restriction to Σ^{rs} (see Proposition 4.5).

Corollary 5.8. *With the above notations:*

(i) *The restricted Grothendieck–Springer sheaf* $\mathcal{G}_\Sigma := \mathrm{R}(\pi_\Sigma)_*\mathbb{Q}_\ell[r+2]$ *is* Λ*-perverse,* Λ*-universally locally acyclic and a relative IC extension of* $\mathcal{L}_\Sigma[r+2] := (\pi^{rs}_\Sigma)_*\mathbb{Q}_\ell[r+2]$.

(ii) *We have an isomorphism* $\mathrm{End}_\Sigma(\mathcal{G}_\Sigma) \xrightarrow{\sim} \mathrm{End}_{\Sigma^{rs}}(\mathcal{L}_\Sigma[r+2])$.

Proof. Note by proper base-change along i_Σ we have $\mathcal{G}_\Sigma = i^*_\Sigma\mathcal{G}[r+2-d]$ where $\mathcal{G}$ is the Grothendieck–Springer sheaf. By the construction of Σ we also have a smooth action map $\alpha : G \times \Sigma \to \mathfrak{g}$ that restricts to smooth action maps $\alpha_{\bar{t}} : G_{\bar{t}} \times \Sigma_{\bar{t}} \to \mathfrak{g}_{\bar{t}}$ along geometric points $\bar{t} \to \Lambda$ (cf. Lemma 4.2 and Proposition 4.3).

As in Proposition 5.7, $\mathcal{G}_\Sigma|_{\Sigma_{\bar{t}}} \simeq \mathrm{R}(\pi_\Sigma)_{\bar{t}*}\mathbb{Q}_\ell[r+2]$ where $(\pi_\Sigma)_{\bar{t}} : \widetilde{\Sigma}_{\bar{t}} \to \Sigma_{\bar{t}}$ is the classical Grothendieck alteration $\pi_{\bar{t}}$ restricted to slice $\Sigma_{\bar{t}}$. We check the smallness of $(\pi_\Sigma)_{\bar{t}}$: put $S_n = \{g \in \mathfrak{g}_{\bar{t}} \mid \dim \pi_{\bar{t}}^{-1}(g) \geq n\}$ then since $\pi_{\bar{t}}$ is a small morphism we have for $n \geq 1$ that $\mathrm{codim}_{\mathfrak{g}_{\bar{t}}}(S_n) > 2n$. As S_n is closed and $G_{\bar{t}}$-invariant and $\alpha_{\bar{t}}$ is smooth, we have

$$\mathrm{codim}_{\Sigma_{\bar{t}}}(\Sigma_{\bar{t}} \cap S_n) = \mathrm{codim}_{G_{\bar{t}} \times \Sigma_{\bar{t}}}(G_{\bar{t}} \times (\Sigma_{\bar{t}} \cap S_n)) = \mathrm{codim}_{G_{\bar{t}} \times \Sigma_{\bar{t}}}(\alpha_{\bar{t}}^{-1}(S_n)) \geq \mathrm{codim}_{\mathfrak{g}_{\bar{t}}}(S_n) > 2n$$

and we deduce that $(\pi_\Sigma)_{\bar{t}}$ is a small morphism as well. Moreover $\widetilde{\Sigma}$ is a smooth Λ-scheme as the composite of smooth maps $\widetilde{\Sigma} \xrightarrow{\widetilde{\chi}_\Sigma} \mathfrak{h} \to \Lambda$ (see Proposition 4.5).

Thus the proof of Proposition 5.7 carries verbatim to this setting as well: $\mathcal{G}_\Sigma$ is Λ-ULA, Λ-perverse and $\mathcal{L}_\Sigma[r+2] = j_\Sigma^* \mathcal{G}_\Sigma$ yields fiberwise $\mathcal{G}_\Sigma|_{\Sigma_{\bar{t}}} = \mathrm{IC}(\mathcal{L}_\Sigma|_{\Sigma_{\bar{t}}})$ so we get (i) and (ii) as before. □

The upshot of Proposition 5.7 and Corollary 5.8 is that the W-action via deck transformations on $(\pi_{\bar{t}}^{\mathrm{rs}})_*\mathbb{Q}_\ell$ (for $\bar{t} \to \Lambda$ a geometric point) naturally and uniquely extends to a W-action on the Grothendieck–Springer sheaf $\mathcal{G}$, which induces the Springer W-action on its stalks at geometric points (cf. Section 5.2). Thus:

Corollary 5.9. *The W-action on the Grothendieck–Springer sheaf $\mathcal{G}$ coming from the W-torsor $\pi^{\mathrm{rs}} : \widetilde{\mathfrak{g}}^{\mathrm{rs}} \to \mathfrak{g}^{\mathrm{rs}}$ induces the Springer W-action on $\mathrm{H}^2(\mathcal{B}_{e_t}, \mathbb{Q}_\ell(1))$ as in* Proposition 5.5*, where $e \to \mathcal{N}$ is fiberwise subregular and $\bar{t} \to \Lambda$ is any geometric point.*

5.4 Monodromy–Springer W-actions. In this section we consider the monodromy I_K-action on the stalks of nearby cycles at a rational double point, and compare it to the Springer W-action as defined in Section 5.3.

Let X/Λ be a surface degeneration with X_k containing a unique rational double point x. For the corresponding $\mathfrak{g}/\Lambda$ and the fiberwise subregular section $e\colon \mathrm{Spec}(\Lambda) \to \mathcal{N}$, construct transverse slice Σ as in Section 4.1 and identify the singularity germ (X_k, x) with germ $(\Sigma_k \cap \mathcal{N}_k, e_k)$. By miniversality of the *completed* family $\widehat{\chi}_\Sigma$ (at e_k in the source and at the origin in the target, see Theorem 4.7), there exists a classifying map with the properties[8]

$$\phi : \mathrm{Spec}(\Lambda) \to \mathfrak{h}/\!\!/W, \quad \phi(\mathrm{Spec}(k)) = 0,\ \phi(\mathrm{Spec}(K)) \in (\mathfrak{h}/\!\!/W)^{\mathrm{rs}}$$

Write $Y := \Sigma \times_\phi \mathrm{Spec}(\Lambda)$, then for $y \in Y_k$ mapping to $e_k \in \Sigma$ we have by construction $\widehat{\mathcal{O}}_{X,x} \simeq \widehat{\mathcal{O}}_{Y,y}$, so Proposition 5.3 yields

$$(\mathrm{R}\Psi_X \mathbb{Q}_\ell)_x \xrightarrow{\sim} (\mathrm{R}\Psi_Y \mathbb{Q}_\ell)_y \tag{5.5}$$

I_K-equivariantly. Note Y_K is smooth, and Y/Λ is nonsmooth only at y.

Set also $Z := \mathrm{Spec}(K) \times_{\mathfrak{h}/\!\!/W} \mathfrak{h}^{\mathrm{rs}} \simeq \mathrm{Spec}(A)$; A is a finite étale K-algebra of dimension $|W|$ since $\psi : \mathfrak{h}^{\mathrm{rs}} \to (\mathfrak{h}/\!\!/W)^{\mathrm{rs}}$ is a W-torsor (Theorem 3.8 (ii)). Note $Z_{\overline{K}} \simeq \coprod_{|W|} \mathrm{Spec}(\overline{K})$ and so a point $a \in Z(\overline{K})$ identifies this set with W.

[8] A priori, formal miniversality gives a map $\widehat{\phi} : \mathrm{Spf}(\Lambda) \to \widehat{\mathfrak{h}/\!\!/W} \simeq \mathrm{Spf}(\Lambda[\![t_1, \cdots, t_r]\!])$, which by restriction along $\Lambda[t_1, \cdots, t_r]$ induces the algebraic map ϕ. Since X_K is smooth, $\phi(\mathrm{Spec}(K))$ lies outside the discriminant locus in $\mathfrak{h}/\!\!/W$, i.e. it lies in $(\mathfrak{h}/\!\!/W)^{\mathrm{rs}}$.

Construction 5.10. We define the following homomorphisms $\rho_a : I_K \to W$ for $a \in Z(\overline{K})$: there is a freely transitive W-action on $Z(\overline{K})$ induced by the geometric Weyl action on $\mathfrak{h}^{\mathrm{rs}}$ via $w \cdot h = \mathrm{Ad}(\widetilde{w})(h)$ for $h \in \mathfrak{h}^{\mathrm{rs}}(\overline{K}), \widetilde{w} \in N_G(T)$ lift of w (Theorem 3.8 (iii)). Denote this action by $w \cdot_1 a \in Z(\overline{K})$. There is another natural action on $Z(\overline{K})$ by I_K, the usual continuous Galois action on the $\overline{K}$-points of a finite étale K-scheme, which we denote by $g \cdot_2 a$. Since G_K and T_K are split, the Weyl action is defined over K, so the two actions commute: $w \cdot_1 (g \cdot_2 a) = g \cdot_2 (w \cdot_1 a)$. Set

$$\rho_a : I_K \to W, \quad g \cdot_2 a = \rho_a(g)^{-1} \cdot_1 a$$

It is straightforward to check ρ_a is a well-defined homomorphism and a different choice of $a \in Z(\overline{K})$ gives a conjugate of ρ_a.

Proposition 5.11. *Let Y, Z and ρ_a be as above. Set $L := (\overline{K})^{\ker \rho_a}$ and let Λ_L be the normalization of Λ in L/K. Then $a \in Z(\overline{K})$ extends to a section $a : \mathrm{Spec}(\Lambda_L) \to \mathfrak{h}$ and the pullback of the restricted Grothendieck–Springer resolution $q : \widetilde{\Sigma} \to \Sigma \times_{\mathfrak{h}/\!/W} \mathfrak{h}$ along a yields a simultaneous resolution*

$$q_a : \widetilde{Y}_a \to Y \times_{\Lambda} \Lambda_L$$

over Λ_L. On the special fibers $q_{a,k}$ is a resolution of Y_k with exceptional divisor $E \simeq \mathcal{B}_{e_k}$.

Proof. Note that L/K is a finite Galois extension of degree $\leq |W|$ since $\ker(\rho_a)$ is an open normal subgroup, and by conjugation L does not depend on a. Since $g \in \ker(\rho_a)$ if and only if $g \cdot_2 a = a$, we have that $a : \mathrm{Spec}(\overline{K}) \to \mathfrak{h}^{\mathrm{rs}}$ factors through $a_L : \mathrm{Spec}(L) \to \mathfrak{h}^{\mathrm{rs}}$, and by the valuative criterion of properness a_L extends to a section $a_{\Lambda_L} : \mathrm{Spec}(\Lambda_L) \to \mathfrak{h}$ with $a_{\Lambda_L}(\mathrm{Spec}(k)) = 0$. Abusing notation, we replace a_{Λ_L} with a.

Set $\widetilde{Y}_a = \widetilde{\Sigma} \times_a \mathrm{Spec}(\Lambda_L)$ and note that the generic fiber $(\widetilde{Y}_a)_L$ is smooth as we base-changed the smooth morphism $\widetilde{\chi}_\Sigma : \widetilde{\Sigma} \to \mathfrak{h}$. By naturality, there is also a canonical identification $\widetilde{Y}_a \simeq \widetilde{\Sigma} \times_{\Sigma \times_{\mathfrak{h}/\!/W} h} (Y \times_\Lambda \Lambda_L)$ and so the universal property of cartesian diagrams yields a map $q_a : \widetilde{Y}_a \to Y \times_\Lambda \Lambda_L$ that is the base-change of map

$$q = (\pi_\Sigma, \widetilde{\chi}_\Sigma) : \widetilde{\Sigma} \to \Sigma \times_{\mathfrak{h}/\!/W} \mathfrak{h}$$

Proposition 4.5 (iii) then yields that q_a is a simultaneous resolution over $\mathrm{Spec}(\Lambda_L)$ (cf. Corollary 3.10 too), i.e. over the generic fiber it is an isomorphism and over the special fiber it is the minimal resolution

$$q_{a,k} \colon (\widetilde{Y}_a)_k \to Y_k \simeq \chi_{\Sigma_k}^{-1}(0) = \Sigma_k \cap \mathcal{N}_k$$

In other words, the exceptional divisor is $E \simeq \mathcal{B}_{e_k}$ (cf. proof of Proposition 5.5). □

Together with the I_K-equivariant isomorphism in (5.5) we have:

Corollary 5.12. *The chosen lift $a : \mathrm{Spec}(\overline{K}) \to \mathfrak{h}^{rs}$ yields an I_L-equivariant isomorphism*

$$\gamma_a : (\mathrm{R}\Psi_X \mathbb{Q}_\ell)_x \xrightarrow{\sim} \mathrm{R}\Gamma(\mathcal{B}_{e_k}, \mathbb{Q}_\ell) \simeq \mathbb{Q}_\ell[0] \oplus V_\ell(-1)[-2] \tag{5.6}$$

where $V_\ell = X^*(T) \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$ and the I_L-action on the source and target is trivial.

Proof. Retaining the notation from Proposition 5.11, set $Y' = Y \times_\Lambda \Lambda_L$ so that $Y'_k = Y_k$. Invariance of geometric nearby cycles under change of trait ([SGA $4\frac{1}{2}$, Th. 3.7]) and Proposition 5.3 give

$$\mathrm{Res}^{I_K}_{I_L}((\mathrm{R}\Psi_X\mathbb{Q}_\ell)_x) \simeq (\mathrm{R}\Psi_{Y'}\mathbb{Q}_\ell)_y$$

By proper base-change and smoothness of $\widetilde{Y}_a/\Lambda_L$, Proposition 5.11 gives I_L-equivariant isomorphisms

$$(\mathrm{R}\Psi_{Y'}\mathbb{Q}_\ell)_y \simeq (\mathrm{R}(q_{a,k})_*\mathrm{R}\Psi_{\widetilde{Y}_a}\mathbb{Q}_\ell)_y \simeq (\mathrm{R}(q_{a,k})_*\mathbb{Q}_\ell)_y \simeq \mathrm{R}\Gamma(\mathcal{B}_{e_k},\mathbb{Q}_\ell)$$

where $q_{a,k} : (\widetilde{Y}_a)_k \to Y_k$ is the minimal resolution and we identify the exceptional divisor with Springer fiber $\mathcal{B}_{e_k}$. The right isomorphism in (5.6) follows from Proposition 5.5. □

To derive a description of the full I_K-action on the right hand side of (5.6), we proceed as follows. Write $\phi_Y : Y \to \Sigma$ for the map induced by ϕ and set $\widetilde{Y} := Y \times_\Sigma \widetilde{\Sigma}$ for the base-change of the restricted Grothendieck alteration π_Σ along ϕ_Y. By the universal property there is a map $\pi_Y : \widetilde{Y} \to Y$ of Λ-schemes, so that over $\overline{K}$ it is $\widetilde{Y}_{\overline{K}} \simeq \coprod_{a\in Z(\overline{K})} Y_{\overline{K}} \to Y_{\overline{K}}$ and over k it is the minimal resolution of Y_k after taking $(\widetilde{Y}_k)_{\mathrm{red}}$ on the source. We have

$$\mathcal{F} := \mathrm{R}(\pi_Y)_*\mathbb{Q}_\ell \simeq \phi_Y^*\mathrm{R}(\pi_\Sigma)_*\mathbb{Q}_\ell = \phi_Y^*\mathcal{G}_\Sigma[-r-2]$$

by proper base-change. Restricting to the special fiber Y_k we have specialization map (Definition 5.1 (ii))

$$\mathcal{F}|_{Y_k} \longrightarrow \mathrm{R}\Psi_Y(\mathcal{F}|_{Y_K})$$

Taking the stalk at $y \in Y$ and noting that $\mathcal{F}_y \simeq \mathrm{R}\Gamma(\mathcal{B}_{e_k},\mathbb{Q}_\ell)$ (Proposition 5.11) and $\mathcal{F}|_{Y_{\overline{K}}} \simeq \mathbb{Q}_{\ell,Y_{\overline{K}}} \otimes \mathbb{Q}_\ell^{Z(\overline{K})}$ (where $Z(\overline{K})$ is a discrete set of cardinality $|W|$) we get a map

$$\mathrm{sp}\colon \mathrm{R}\Gamma(\mathcal{B}_{e_k},\mathbb{Q}_\ell) \longrightarrow (\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y \otimes_{\mathbb{Q}_\ell} \mathbb{Q}_\ell^{Z(\overline{K})}$$

For the following lemma, we recall the actions of W and I_K on $Z(\overline{K})$ as in Construction 5.10. They respectively induce a W-action and an I_K-action on $(\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y \otimes_{\mathbb{Q}_\ell} \mathbb{Q}_\ell^{Z(\overline{K})}$ as follows: after identifying the latter with $\prod_{a\in Z(\overline{K})}(\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y$ and indexing elements by v_a, the W-action is $w\cdot(v_a)_a := (v_{w^{-1}\cdot 1 a})_a$ and the I_K-action is $g\cdot(v_a)_a := (g(v_{g^{-1}\cdot 2 a}))_a$ where $g \in I_K$ acts as the monodromy on the right hand side.

Lemma 5.13. *With notations as above:*

(i) *Specialization map* sp *is W-equivariant and I_K-equivariant, where the source is equipped with the Springer W-action from* Corollary 5.9 *and the trivial I_K-action, and the target is equipped with the W-action and the I_K-action specified above.*

(ii) *Choose $a \in Z(\overline{K})$ and set*

$$\delta_a = (\mathrm{id}_{\mathrm{R}\Psi} \otimes \mathrm{ev}_a) \circ \mathrm{sp}\colon \mathrm{R}\Gamma(\mathcal{B}_{e_k},\mathbb{Q}_\ell) \longrightarrow (\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y \otimes_{\mathbb{Q}_\ell} \mathbb{Q}_\ell^{Z(\overline{K})} \longrightarrow (\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y$$

where ev_a *specifies a component of* $\mathbb{Q}_\ell^{Z(\overline{K})} \simeq \mathbb{Q}_\ell^{|W|}$. *Then δ_a is the inverse of isomorphism γ_a from* Corollary 5.12.

Proof. For (i), I_K-equivariance follows from the naturality of the specialization (Definition 5.1) and W-equivariance follows from the W-action on $\mathcal{F}$ induced from Corollary 5.8.

For (ii) we implicitly identify $(\mathrm{R}\Psi_X\mathbb{Q}_\ell)_x \simeq (\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y$ as in the proof of Corollary 5.12. By Proposition 5.11, $a \in Z(\overline{K})$ extends to a section $a : \mathrm{Spec}(\Lambda_L) \to \mathfrak{h}$ where L is the fixed field of $\ker \rho_a \subset I_K$. Over $Y \times_\Lambda \Lambda_L$ we have an inclusion $\widetilde{Y}_a \coloneqq \widetilde{\Sigma} \times_a \mathrm{Spec}(\Lambda_L) \hookrightarrow \widetilde{Y} \times_\Lambda \Lambda_L$ and adjunction gives

$$\mathcal{F}|_{Y\times_\Lambda\Lambda_L} \longrightarrow \mathrm{R}(q_a)_*\mathbb{Q}_\ell, \quad q_a : \widetilde{Y}_a \to Y \times_\Lambda \Lambda_L$$

where q_a is the simultaneous resolution of Proposition 5.11. Over $Y_{\overline{K}}$ this map is just ev_a and over Y_k it is an isomorphism as $\pi_{Y,k}$ and $q_{a,k}$ are the same minimal resolutions after passing to the reduced subscheme structure. The specialization map for $\mathrm{R}(q_a)_*\mathbb{Q}_\ell|_{Y_k} \simeq \mathrm{R}(q_{a,k})_*\mathbb{Q}_\ell$ is

$$\mathrm{sp}' : \mathrm{R}(q_{a,k})_*\mathbb{Q}_\ell \longrightarrow \mathrm{R}\Psi_{Y'}(\mathrm{R}(q_a)_*\mathbb{Q}_\ell|_{Y_{\overline{K}}}) \simeq \mathrm{R}\Psi_{Y'}\mathbb{Q}_\ell \simeq \mathrm{Res}^{I_K}_{I_L}\mathrm{R}\Psi_Y\mathbb{Q}_\ell$$

as q_a is an isomorphism over $\mathrm{Spec}(\overline{K})$. By naturality of specialization maps, after taking stalks at y we have

$$\begin{array}{ccc} \mathrm{R}\Gamma(\mathcal{B}_{e_k},\mathbb{Q}_\ell) & \xrightarrow{\mathrm{sp}} & (\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y \otimes_{\mathbb{Q}_\ell} \mathbb{Q}_\ell^{Z(\overline{K})} \\ \downarrow\simeq & & \downarrow{\scriptstyle \mathrm{id}\otimes\mathrm{ev}_a} \\ (\mathrm{R}(q_a)_*\mathbb{Q}_\ell)_y & \xrightarrow{\mathrm{sp}'_y} & (\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y \end{array}$$

The left column and bottom row of the diagram define γ_a^{-1} while the top row and right column define δ_a, which are then the same by commutativity. □

Proposition 5.14. *Let $\mathrm{H}^2(\gamma_a) : (\mathrm{R}^2\Psi_Y\mathbb{Q}_\ell)_y \to \mathrm{H}^2(\mathcal{B}_{e_k},\mathbb{Q}_\ell)$ be the isomorphism induced from γ_a, then it identifies the monodromy I_K-action with the Springer W-action via $\rho_a : I_K \to W$ in the sense that*

$$\mathrm{H}^2(\gamma_a)(g\cdot v) = \rho_a(g)\cdot \mathrm{H}^2(\gamma_a)(v), \quad g \in I_K, v \in (R^2\Psi_Y\mathbb{Q}_\ell)_y \tag{5.7}$$

where $g\cdot(-)$ denotes the monodromy action and $\rho_a(g)\cdot(-)$ denotes the Springer action.

Proof. We first explain two compatibility relations on maps

$$\mathrm{H}^2(\delta_b) : \mathrm{H}^2(\mathcal{B}_{e_k},\mathbb{Q}_\ell) \to (\mathrm{R}^2\Psi_Y\mathbb{Q}_\ell)_y, \quad b \in Z(\overline{K})$$

with the understanding that δ_b maps to the copy of $(\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y$ indexed by b. By I_K-equivariance of the specialization map (where I_K acts trivially on the source, see Lemma 5.13),

$$g\cdot \mathrm{H}^2(\delta_{g^{-1}\cdot_2 b})(v) = \mathrm{H}^2(\delta_b)(v), \quad v \in \mathrm{H}^2(\mathcal{B}_{e_k},\mathbb{Q}_\ell)$$

so setting $a = g^{-1}\cdot_2 b$ yields $g\circ\delta_a = \delta_{g\cdot_2 a}$ for the monodromy I_K-action on $(\mathrm{R}\Psi_Y\mathbb{Q}_\ell)_y$. Similarly, W-equivariance of the specialization map from Lemma 5.13 yields relation $\delta_a \circ w = \delta_{w^{-1}\cdot_1 a}$. Thus

$$g\circ\delta_a = \delta_{g\cdot_2 a} = \delta_{\rho_a(g)^{-1}\cdot_1 a} = \delta_a\circ\rho_a(g)$$

by definition of ρ_a. Since $\delta_a = \gamma_a^{-1}$ by Lemma 5.13, we obtain $\mathrm{H}^2(\gamma_a) \circ g = \rho_a(g) \circ \mathrm{H}^2(\gamma_a)$, which yields (5.7). □

Along with the I_K-equivariant isomorphism $(\mathrm{R}^2\Psi_X\mathbb{Q}_\ell)_x \simeq (\mathrm{R}^2\Psi_Y\mathbb{Q}_\ell)_y$, the upshot is that the natural monodromy I_K-action on the left-hand side is exactly the restriction of the Springer W-action along a map $\rho_a : I_K \to W$. By Corollary 5.12 $I_L = \ker \rho_a$ acts trivially on $(\mathrm{R}\Psi_X\mathbb{Q}_\ell)_x$ so ρ_a factors through

$$\overline{\rho}_a : I_K/I_L \simeq \mathrm{Gal}(L/K) \hookrightarrow W$$

By definition of ρ_a, picking another element $b = w \cdot_1 a \in Z(\overline{K})$ yields $\overline{\rho}_b(g) = w\overline{\rho}_a(g)w^{-1}$ and so ρ_a, ρ_b realize $\mathrm{Gal}(L/K)$ as conjugate subgroups in W. In other words:

Corollary 5.15. *The monodromy action on $(R^2\Psi_X\mathbb{Q}_\ell)_x$ factors through an embedding $\overline{\rho}_a : \mathrm{Gal}(L/K) \to W$ and is the restriction of Springer's reflection W-representation to a subgroup whose conjugacy class depends only on $\phi : \mathrm{Spec}(\Lambda) \to \mathfrak{h}/\!\!/W$.*

5.5 Proof of the main theorem. We can finally put the pieces together in order to prove Theorem 1.1, re-phrased below. We retain the notations of Section 5.1, 5.2 and 5.4.

Theorem 5.16. *Let X/Λ be a surface degeneration with X_k having a unique rational double point x so that p is good for x. Let Φ, W be the root lattice associated to x and its Weyl group, and set $V_\ell = \Phi^\vee \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$.*

(i) *The monodromy I_K-action on $\mathrm{H}^i(X_{\overline{K}}, \mathbb{Q}_\ell)$ is trivial for $i \neq 2$. For $i = 2$ let L be the fixed field of the kernel of the I_K-action, then we have an I_K-equivariant splitting*

$$\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^2(X_k, \mathbb{Q}_\ell) \oplus V_\ell(-1) \tag{5.8}$$

The monodromy I_K-action is trivial on the first summand, and on the second it factors through $\mathrm{Gal}(L/K)$, where it is identified with the restriction of the reflection W-representation along an embedding $\mathrm{Gal}(L/K) \hookrightarrow W$.

(ii) *L is the minimal Galois extension over which X_K has good reduction in the category of algebraic spaces. Moreover the isomorphism class of the I_K-representation $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ depends, up to trivial summands, only on a formal Λ-isomorphism class of (X, x). In particular L depends only on this class.*

Remark 5.17. We make several comments on the formulation of Theorem 5.16.

(i) A uniform characteristic that works is $p > 5$; the bound can be lower if we know the Dynkin type of x. Moreover, the local monodromy I_K-action on V_ℓ and extension L are invariant under passage to an étale neighborhood of (X, x).

(ii) The 'up to trivial summands' condition is important: blowing up a section of X disjoint from x contributes a trivial I_K-representation $\mathbb{Q}_\ell(-1)$ as a summand in $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ without changing the nontrivial part of the monodromy; in particular, if we fix $\dim \mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ we have that a formal Λ-germ of x determines the monodromy representation up to isomorphism.

(iii) The statement generalizes to X_k having rational double points $x_1, \cdots, x_m$. If Φ_i is the Dynkin type of x_i with Weyl group W_i and V_i is the $\mathbb{Q}_\ell$-vector space reflection representation of W_i, set $W = \prod W_i$ and $V_\ell = \bigoplus V_i$. Assuming p is good for each Φ_i, we have an I_K-equivariant splitting $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^2(X_k, \mathbb{Q}_\ell) \oplus V_\ell(-1)$ with the monodromy action being trivial on the first summand, while on the second summand it is the restriction of the reflection representation of W along

$$I_K \to \mathrm{Gal}(L/K) \hookrightarrow W$$

Here L is the compositum of L_i where L_i is the monodromy extension associated to x_i as in (i).

Before the proof, we construct a smooth model for X after base extension. Recall the homomorphism $\rho_a : I_K \to W$ (Construction 5.10) which yields a finite Galois extension L/K as in Proposition 5.11.

Lemma 5.18. *There exists a proper morphism of algebraic spaces $\widetilde{X} \to X \times_\Lambda \Lambda_L$ with $\widetilde{X}$ smooth over Λ_L. The map is an isomorphism outside of x and over the special fiber $\widetilde{X}_k \to X_k$ is a minimal resolution of X_k.*

Proof. Take (Y, y) as in the setup of Section 5.4. By Proposition 5.3 there is a common pointed étale neighborhood

$$(X, x) \leftarrow (U, u) \to (Y, y)$$

and up to shrinking we can assume the preimage of x and y is just u. Pull back the simultaneous resolution $q_a : \widetilde{Y}_a \to Y \times_\Lambda \Lambda_L$ of Proposition 5.11 to $\widetilde{U} \to U \times_\Lambda \Lambda_L$; $\widetilde{U}$ is étale over $\widetilde{Y}_a$ hence smooth over Λ_L, and this modification is an isomorphism outside u. By [Stacks, Tag 0BGY] we can descend this modification along $U \times_\Lambda \Lambda_L \to X \times_\Lambda \Lambda_L$ and glue it with the identity modification outside x to obtain a proper modification $\widetilde{X} \to X \times_\Lambda \Lambda_L$. By construction $\widetilde{X}$ is a proper smooth algebraic space, and since the special fiber is the minimal resolution over the pointed étale neighborhood (U, u) and the modification is the identity otherwise, $\widetilde{X}_k$ is the minimal resolution of X_k. □

We also need to consider separately the case of p good but not very good, i.e. $p \mid n$ when x is an A_{n-1}-singularity; the upshot here is that we can repeat the slice arguments with an appropriate slice inside $\mathfrak{gl}_n$ instead of $\mathfrak{sl}_n$.

Lemma 5.19. *In the setting of* Theorem 5.16, *suppose $p \mid n$ and $x \in X_k$ is an A_{n-1} singularity. Let $\mathfrak{g} = \mathfrak{gl}_n = \mathrm{Lie}(\mathrm{GL}_{n,\Lambda})$, $\mathfrak{h}$ the diagonal Cartan algebra, $W = S_n$ and $V_\ell = \mathbb{Q}_\ell^n / \mathbb{Q}_\ell(1, \cdots, 1)$ the quotient by the diagonal embedding $\mathbb{Q}_\ell \hookrightarrow \mathbb{Q}_\ell^n$.*

(i) *There exists a homogeneous transverse slice Σ at a fiberwise subregular section such that $\chi_\Sigma : \Sigma \to \mathfrak{h}/\!/W$ induces a formal miniversal deformation of the A_{n-1} singularity, and moreover $\widetilde{\Sigma} \to \mathfrak{h}$ is smooth and $\widetilde{\Sigma} \to \Sigma \times_{\mathfrak{h}/\!/W} \mathfrak{h}$ is an isomorphism over the regular semisimple locus and yields a minimal resolution of $\chi_\Sigma^{-1}(0)$.*

(ii) *We have an isomorphism $(\mathrm{R}^2\Psi_X\mathbb{Q}_\ell)_x \simeq V_\ell(-1)$ where the I_K-action on the right is a restriction of the reflection representation along a map $\rho_a : I_K \to W$.*

Proof. We first explain (i). Let $e = B_{n-1} \oplus 0$ where B_{n-1} is the $(n-1) \times (n-1)$ matrix with 1 along its superdiagonal and 0 in all other entries. Fiberwise its Jordan type is $(n-1, 1)$ so e is fiberwise subregular nilpotent. Let

$$\Sigma := \{e + \sum_{i=1}^{n-1} a_i E_{n-1,i} + xE_{n-1,n} - yE_{n,1} + zE_{n,n}\} \simeq \mathbb{A}^{n+2}_\Lambda$$

Computing the characteristic polynomial of matrices in Σ, we get

$$\Sigma \simeq \mathrm{Spec}\left(\frac{\Lambda[x, y, z, t_1, \cdots, t_n]}{xy - z^n + t_1 z^{n-1} + \cdots + t_n}\right) \tag{5.9}$$

Slice Σ is the relative version of the matrix slice constructed in [MV22, §3.2.4] for Jordan type $(n-1, 1)$, and after completing the map $\Sigma \to \mathfrak{h}/\!/S_n \simeq \mathrm{Spec}(\Lambda[t_1, \cdots, t_n])$ it yields a formal miniversal deformation for singularity A_{n-1} (see Proposition 2.18 and algebra B in Example 2.19).

Put $\mathfrak{a} = \Sigma - e$. A matrix computation shows that the trace pairing $(A, B) \mapsto \mathrm{tr}(AB)$ restricts to a perfect pairing between $\mathfrak{a}$ and centralizer $Z_\mathfrak{g}(e)$ so $\mathfrak{g} = [\mathfrak{g}, e] \oplus \mathfrak{a}$. The proof of Proposition 4.3 therefore applies to show Σ is a homogeneous transverse slice at e.

By the proof of Theorem 3.8 (i) which applies to $\mathfrak{g} = \mathfrak{gl}_n$, we have that $\widetilde{\chi} : \widetilde{\mathfrak{g}} \to \mathfrak{h}$ is smooth and so restricting the Grothendieck alteration to Σ as usual shows that $\widetilde{\Sigma} \to \mathfrak{h}$ is smooth. The last statement follows from resolution $(\widetilde{\Sigma}_k)_0 \to \chi_{\Sigma_k}^{-1}(0)$ having a chain of $n-1$ rational (-2) curves as exceptional divisor, proving minimality.

For (ii), we first identify V_ℓ: the Chern class argument of Proposition 5.5 gives an S_n-equivariant map $\mathbb{Q}_\ell^n \to \mathrm{H}^2(\mathcal{B}_{e_k}, \mathbb{Q}_\ell(1))$ which has coordinates $(\lambda_i - \lambda_{i+1})_i$. The diagonal embedding then gives $V_\ell \simeq \mathbb{Q}_\ell^n/\mathbb{Q}_\ell(1, \cdots, 1)$.

Now $\widehat{\mathcal{O}}_{X,x}$ is a formal deformation of the A_{n-1}-singularity over $\mathrm{Spec}(\Lambda)$ so there exist ϕ, Y, y as in Section 5.4 so that $(\mathrm{R}^2\Psi_X\mathbb{Q}_\ell)_x \simeq (\mathrm{R}^2\Psi_Y\mathbb{Q}_\ell)_y$ by Proposition 5.3. The arguments of Proposition 5.11, Corollary 5.12, Lemma 5.13 and Proposition 5.14 apply verbatim to show $(\mathrm{R}^2\Psi_Y\mathbb{Q}_\ell)_y \simeq V_\ell(-1)$. □

Proof of Theorem 5.16. We first assume p is very good. Since X/Λ is proper we have $\mathrm{H}^2(X_k, \mathrm{R}\Psi_X\mathbb{Q}_\ell) \simeq H^2(X_{\overline{K}}, \mathbb{Q}_\ell)$, so the I_K-equivariant triangle in Definition 5.1 (ii) gives

$$0 \to \mathrm{H}^2(X_k, \mathbb{Q}_\ell) \to \mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell) \xrightarrow{(*)} (\mathrm{R}^2\Psi_X\mathbb{Q}_\ell)_x \to \mathrm{H}^3(X_k, \mathbb{Q}_\ell) \to \mathrm{H}^3(X_{\overline{K}}, \mathbb{Q}_\ell) \to 0 \tag{5.10}$$

since $(\mathrm{R}^i\Psi_X\mathbb{Q}_\ell)_x = 0$ for $i \neq 0, 2$. To check surjectivity of $(*)$, let $\widetilde{X} \to X \times_\Lambda \Lambda_L$ be a simultaneous resolution in the category of algebraic spaces over Λ_L as in Lemma 5.18.

Then $(*)$ is identified with restriction $i^* : \mathrm{H}^2(\widetilde{X}_k, \mathbb{Q}_\ell) \to \mathrm{H}^2(\mathcal{B}_{e_k}, \mathbb{Q}_\ell)$ where $i : \mathcal{B}_{e_k} \hookrightarrow \widetilde{X}_k$ is identified with the exceptional divisor of minimal resolution $\widetilde{X}_k$. Note $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^2(\widetilde{X}_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^2(\widetilde{X}_k, \mathbb{Q}_\ell)$, the latter because $\widetilde{X}$ is smooth.

Label the exceptional curves in $\widetilde{X}_k$ by $E_1, \cdots, E_r$ where $r = \mathrm{rk}(\Phi)$. By Proposition 5.5, restricting $\mathcal{B}_{e_k} = \cup_i E_i$ to its irreducible components gives

$$\mathrm{H}^2(\mathcal{B}_{e_k}, \mathbb{Q}_\ell(1)) \simeq \bigoplus_{i=1}^{r} \mathrm{H}^2(E_i, \mathbb{Q}_\ell(1)) \simeq V_\ell \tag{5.11}$$

Let $e_i = c_1(\mathcal{O}(E_i)) \in \mathrm{H}^2(\widetilde{X}_k, \mathbb{Q}_\ell(1))$ denote the Chern classes of E_i, then the identification in (5.11) sends $i^* e_j \mapsto (\deg(\mathcal{O}(E_j)|_{E_i}))_i = (E_j \cdot E_i)_i$ so overall we have a linear map

$$V_\ell \to \mathrm{H}^2(\widetilde{X}_k, \mathbb{Q}_\ell(1)) \xrightarrow{i^*} \mathrm{H}^2(\mathcal{B}_{e_k}, \mathbb{Q}_\ell(1)) \simeq V_\ell$$

where the leftmost map is $(a_i) \mapsto (\sum a_i e_i)$. The matrix of the composite map is the (negative of the) Cartan matrix, whose determinant is nonzero and so i^* is surjective. We deduce from this that $\mathrm{H}^i(X_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^i(X_k, \mathbb{Q}_\ell)$ is a trivial I_K-module for $i \neq 2$.

Smooth proper base-change for $\widetilde{X}$ shows that $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ is a trivial I_L-module, so (5.10) is a sequence of $\mathrm{Gal}(L/K)$-representations i.e. $(*)$ splits and we get (5.8) I_K-equivariantly. This yields (i) for p very good. The remaining case is when x is an A_n-singularity and $p \mid n+1$, which is treated in Lemma 5.19.

For (ii), note that by Proposition 5.3 the I_K-representation $(\mathrm{R}^2\Psi_X \mathbb{Q}_\ell)_x$ depends only on a formal affine neighborhood of (X, x) over Λ, hence so does $\ker \rho_a \subset I_K$ and $L = \overline{K}^{\ker \rho_a}$.

It remains to show the minimality of L. Let M/K be a finite extension and suppose X_M admits a smooth proper model over Λ_M (the normalization of Λ inside M). Smooth proper base-change implies I_M acts trivially on $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ so $I_M \subset \ker \rho_a = I_L$; as k is algebraically closed this implies $L \subset M$. Conversely smooth proper model $\widetilde{X}$ over Λ_L base-changes to a smooth proper model over Λ_M. We conclude L/K is the minimal extension over which X_K has good reduction in the category of algebraic spaces. □

Remark 5.20. Lemma 5.18 produces a smooth proper algebraic space model for X after a finite base extension. We do *not* in general get that X has potential good reduction in the category of schemes, so it is necessary to enlarge to the category of algebraic spaces. In [Mat15, §5.2], Matsumoto constructs a surface degeneration

$$X = \{w^2 = f(x, y, z)\} \subset \mathbb{P}_\Lambda(1, 1, 1, 3)$$

where $f(x, y, z)$ is a homogeneous sextic polynomial, $\Lambda = W(k)$ and $\mathrm{char}(k) = p \geq 7$, so that X_K is a smooth K3 surface of geometric Picard rank 1 and special fiber X_k has an ordinary double point. By [Mat15, Prop. 5.1] X_K does not admit a smooth proper scheme model over any finite extension.

When p is sufficiently good (see Definition 2.3 (iii)) we have $p \nmid |\mathrm{Gal}(L/K)|$ so pro-p wild inertia P_K acts trivially, L/K is a tame extension and $\mathrm{Gal}(L/K) = \langle T \rangle$ up to conjugacy. So as a special case of Theorem 5.16 we have:

Corollary 5.21. *For p sufficiently good, the ℓ-adic monodromy is generated by a single element $T \in W$ of order $|\mathrm{Gal}(L/K)|$, determined up to conjugacy and acting on the $V_\ell = \Phi^\vee \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$ summand of $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell(1))$ through the reflection representation.*

6. The case of general rational singularities

In this section, we explain how to extend Theorem 1.1 to the case where X/Λ is a surface degeneration with X_k having a unique rational singularity $x \in X(k)$. We retain the hypotheses of Section 5.1 regarding Λ.

Write $R = \widehat{\mathcal{O}}_{X_k,x}$ and recall that R admits formal miniversal deformations which admit algebraic representatives (see Section 2.5). Let $A \in \mathrm{Comp}_{\Lambda,k}$ denote the base of a formal miniversal deformation of R, then there exists an algebraic classifying map $\phi : \mathrm{Spec}(\Lambda) \to \mathrm{Spec}(A)$ induced from deformation $\widehat{\mathcal{O}}_{X,x}$ of R. We will throughout assume the following:

Assumption $(*)$**.** The classifying map ϕ factors as $\mathrm{Spec}(\Lambda) \to \mathrm{Spec}(A/I) \to \mathrm{Spec}(A)$ where $\mathrm{Spec}(A/I)$ is the Artin component of $\mathrm{Spec}(A)$ (Definition 2.27).

Lemma 6.1. *Suppose X/Λ induces a formal deformation of R with classifying map ϕ satisfying $(*)$. Then there exists a finite extension Λ_L of Λ such that the base-change $X' \coloneqq X \times_{\mathrm{Spec}(\Lambda)} \mathrm{Spec}(\Lambda_L)$ admits a simultaneous resolution $\widetilde{X} \to X'$ over Λ_L.*

Proof. By Proposition 2.28 there exists a finite A-algebra C that is formally smooth over Λ and such that the algebraic realization $\mathcal{U} \to \mathrm{Spec}(A)$ of the miniversal deformation of R admits a simultaneous resolution after base-change $\mathrm{Spec}(C) \to \mathrm{Spec}(A)$.

Let $\pi : \widetilde{\mathcal{U}} \to \mathcal{U}_C \coloneqq \mathcal{U} \times_{\mathrm{Spec}(A)} \mathrm{Spec}(C)$ be a simultaneous resolution and put $\Lambda' = C \otimes_{A/I} \Lambda$, so $\mathrm{Spec}(\Lambda') \to \mathrm{Spec}(\Lambda)$ is finite surjective and in particular $\Lambda' \otimes_\Lambda K \neq 0$. Choose a closed point of $\mathrm{Spec}(\Lambda' \otimes_\Lambda K)$ which corresponds to a finite extension L/K, and let Λ_L be the normalization of Λ in L, which is also a complete discrete valuation ring with residue field k.

Write $\mathcal{U}_\Lambda \coloneqq \mathcal{U} \times_{\mathrm{Spec}(A)} \mathrm{Spec}(\Lambda)$. By construction there is a point $u \in (\mathcal{U}_\Lambda)_k$ such that $\widehat{\mathcal{O}}_{X,x} \simeq \widehat{\mathcal{O}}_{\mathcal{U}_\Lambda,u}$ and by the proof of Proposition 5.3 we can choose a pointed étale neighborhood (T,t) of both (X,x) and $(\mathcal{U}_\Lambda, u)$. Pulling back the simultaneous resolution π to $\mathrm{Spec}(\Lambda_L)$ we get a simultaneous resolution $\pi_\Lambda : \widetilde{\mathcal{U}}_{\Lambda_L} \to \mathcal{U}_{\Lambda_L}$ over Λ_L. The modification gluing argument of Lemma 5.18 applied to π_Λ and étale neighborhood (T,t) yields a modification $\widetilde{X} \to X'$ that is an isomorphism over $\mathrm{Spec}(L)$ and the minimal resolution of X_k over $\mathrm{Spec}(k)$. □

Proposition 6.2. *Suppose X/Λ admits a simultaneous resolution after a finite extension L/K. Then there exists a simultaneous partial resolution $f : X^{\mathrm{RDP}} \to X$ over Λ in the sense that f is proper with $f|_{\mathrm{Spec}(K)}$ an isomorphism, X^{RDP} is flat over Λ and $f|_{\mathrm{Spec}(k)}$ is a partial resolution of X_k, obtained by contracting the (-2)-curves in its minimal resolution.*

Proof. Set $\mathcal{L} \coloneqq \omega_{\widetilde{X}/\Lambda_L}$ for the relative dualizing sheaf of the simultaneous resolution $\widetilde{X}$ of $X' = X \times_{\mathrm{Spec}(\Lambda)} \mathrm{Spec}(\Lambda_L)$ and denote the exceptional irreducible curves in the minimal resolution $\widetilde{X}_k$ by $E_1, \cdots E_r$. By adjunction $\deg(\mathcal{L}|_{E_i}) = \deg(\omega_{E_i}) - E_i^2 = -2 - E_i^2 \geq 0$,

the last inequality following from $\widetilde{X}_k$ being a minimal resolution of a rational singularity, and thus $E_i^2 \leq -2$ ([Lip69, Cor. 27.3]).

Write $\pi_k : \widetilde{X}_k \to X_k$ for the minimal resolution and note $\mathrm{R}^1(\pi_k)_*\mathcal{O}_{\widetilde{X}_k} = 0$ as $x \in X(k)$ is a rational singularity. We localize by taking $S = \mathrm{Spec}(\mathcal{O}_{X_k,x})$, $g\colon T = S \times_{X_k} \widetilde{X}_k \to S$ so that $\mathrm{R}^1 g_*\mathcal{O}_T = 0$ by flat base-change. As the fibers of g have dimension ≤ 1 and S is affine, by the Leray spectral sequence we get $\mathrm{H}^1(T, \mathcal{O}_T) = 0$, so applying [Lip69, Thm. 12.1 (ii)] we get that $\mathcal{L}^n|_T$ is globally generated and $\mathrm{H}^1(T, \mathcal{L}^n|_T) = 0$ for all $n \geq 1$. Since π_k is an isomorphism outside of x, we get that $\mathcal{L}^n|_{\widetilde{X}_k}$ is relatively globally generated in the sense that $(\pi_k)^*(\pi_k)_*\mathcal{L}^n|_{\widetilde{X}_k} \twoheadrightarrow \mathcal{L}^n|_{\widetilde{X}_k}$ and $\mathrm{R}^1(\pi_k)_*\mathcal{L}^n|_{\widetilde{X}_k} = 0$.

Write $\pi : \widetilde{X} \to X'$ for the simultaneous resolution and let $i : \widetilde{X}_k \hookrightarrow \widetilde{X}$. If $\varpi \in \Lambda_L$ is a uniformizer, applying π_* to

$$0 \to \mathcal{L}^n \xrightarrow{\cdot\varpi} \mathcal{L}^n \to i_*\left(\mathcal{L}^n|_{\widetilde{X}_k}\right) \to 0$$

yields exact sequence

$$\cdots \to \mathrm{R}^1\pi_*\mathcal{L}^n \xrightarrow{\cdot\varpi} \mathrm{R}^1\pi_*\mathcal{L}^n \to j_*\mathrm{R}^1(\pi_k)_*\mathcal{L}^n|_{\widetilde{X}_k} = 0, \quad \text{where } j : X_k \hookrightarrow X' \tag{6.1}$$

Note that for any $y \in X_k$, setting $M = (\mathrm{R}^1\pi_*\mathcal{L}^n)_y$ yields $M = \varpi M$ by (6.1) so $M = 0$ by Nakayama. Since π is an isomorphism over $X' \setminus X_k$ we get overall that $\mathrm{R}^1\pi_*\mathcal{L}^n = 0$ and (6.1) yields $\pi_*\mathcal{L}^n/\varpi(\pi_*\mathcal{L}^n) \simeq j_*(\pi_k)_*\mathcal{L}^n|_{\widetilde{X}_k}$. Setting $\mathcal{F} \coloneqq \mathrm{coker}(\pi^*\pi_*\mathcal{L}^n \to \mathcal{L}^n)$ for the cokernel of the evaluation map, we have $i^*\mathcal{F} = \mathrm{coker}((\pi_k)^*(\pi_k)_*\mathcal{L}^n|_{\widetilde{X}_k} \to \mathcal{L}^n|_{\widetilde{X}_k})$ by proper base-change, and this cokernel is zero as $\mathcal{L}^n|_{\widetilde{X}_k}$ is globally generated. Since $\mathcal{F}|_{\widetilde{X}\setminus\widetilde{X}_k} = 0$ we have $\mathcal{F} = 0$, i.e. $\pi^*\pi_*\mathcal{L}^n \twoheadrightarrow \mathcal{L}^n$, so $\mathcal{L}^n$ is (relatively) globally generated.

Set $\mathcal{A} = \bigoplus_{n\geq 0} \pi_*\mathcal{L}^n$ and $Y = \mathrm{Proj}_{X'}\mathcal{A}$. By the argument above, the graded $\mathcal{O}_{\widetilde{X}}$-algebra map $\pi^*\mathcal{A} \to \bigoplus_{n\geq 0} \mathcal{L}^n$ is surjective in degree 1 and $\mathcal{A}$ is generated in degree 1, so by [Stacks, Tag 085D] the simultaneous resolution $\pi : \widetilde{X} \to X'$ factors as

$$\widetilde{X} \xrightarrow{\alpha} Y \xrightarrow{\beta} X'$$

with α proper, $\alpha^*\mathcal{O}_Y(1) \simeq \mathcal{L}$ and β projective. By construction Y is flat over Λ_L and its generic and special fibers are respectively

$$Y_L \simeq \mathrm{Proj}_{X_L} \bigoplus_{n\geq 0} \omega_{X_L}^n = X_L, \quad Y_k \simeq \mathrm{Proj}_{X_k} \bigoplus_{n\geq 0} (\pi_k)_*\omega_{\widetilde{X}_k}^n$$

Let $\widetilde{X}_k \xrightarrow{h} Y_k' \xrightarrow{\rho} X_k$ be a factorization of the minimal resolution so that $h : \widetilde{X}_k \to Y_k'$ is a contraction of the exceptional (-2)-curves; Y_k' is a partial resolution of X_k which has only rational double point singularities and $h^*\omega_{Y_k'} \simeq \omega_{\widetilde{X}_k}$ with $\omega_{Y_k'}$ ρ-ample ([Lip79, §1 and §3, Lem. 2]). Thus by relative ampleness we get

$$Y_k \simeq \mathrm{Proj}_{X_k} \bigoplus_{n\geq 0} (\pi_k)_*\omega_{\widetilde{X}_k}^n \simeq \mathrm{Proj}_{X_k} \bigoplus_{n\geq 0} \rho_*\omega_{Y_k'}^n \simeq Y_k'$$

so Y_k is a partial resolution of X_k with at worst RDP singularities.

To obtain a model over Λ, let U be the relative smooth locus of X/Λ and let $P = \mathrm{Proj}_X\left(\bigoplus_{n\geq 0}\mathrm{Sym}^n_{\mathcal{O}_X}(\omega_{X/\Lambda})\right)$ so that $P|_U \simeq U$ (as $\omega_{X/\Lambda}$ is invertible over U). Write $\overline{U}$ for the scheme-theoretic closure of $U \hookrightarrow P$. Since projectivization commutes with base-change,

$$P' := P \times_{\mathrm{Spec}(\Lambda)} \mathrm{Spec}(\Lambda_L) \simeq \mathrm{Proj}_{X'}\left(\bigoplus_{n\geq 0}\mathrm{Sym}^n_{\mathcal{O}_{X'}}(\omega_{X'/\Lambda_L})\right)$$

and the schematic closure of $U \times_{\mathrm{Spec}(\Lambda)} \Lambda_L$ in P' is identified with Y by [Lip79, §2]. Set $X^{\mathrm{RDP}} := \overline{U}$ and f the projective composite map $\overline{U} \hookrightarrow P \to X$, then by the above we have $X^{\mathrm{RDP}} \times_{\mathrm{Spec}(\Lambda)} \mathrm{Spec}(\Lambda_L) \simeq Y$. Note $X^{\mathrm{RDP}}_K \simeq X_K$, $X^{\mathrm{RDP}}_k \simeq Y_k$ and X^{RDP} is flat over Λ since Y is flat over Λ_L and $\mathrm{Spec}(\Lambda_L) \to \mathrm{Spec}(\Lambda)$ is faithfully flat, so X^{RDP}/Λ is the desired model with a partial simultaneous resolution f. □

Note that X^{RDP}_K is smooth and X^{RDP}_k contains only rational double points. We can therefore apply the arguments of Theorem 5.16 and deduce the following.

Theorem 6.3. *Let X/Λ be a surface degeneration with X_k having a rational singularity $x \in X(k)$ so that the induced formal deformation of (X_k, x) satisfies* Assumption $(*)$. *Let*

$$\Psi = \bigoplus_{E_i^2=-2} \mathbb{Z}[E_i] \subset \Phi = \bigoplus_{i=1}^{r} \mathbb{Z}[E_i]$$

be an inclusion of exceptional lattices, where E_i are the exceptional curves of the minimal resolution $\widetilde{X}_k \to X_k$. Ψ is a direct sum of finitely many connected ADE root lattices and its Weyl group W acts on Φ via the standard reflections. If p is good for each ADE root lattice in Ψ then

(i) *The monodromy I_K-action on $\mathrm{H}^i(X_{\overline{K}}, \mathbb{Q}_\ell)$ is trivial for $i \neq 2$. For $i = 2$ let L be the fixed field of the kernel of the I_K-action, then there exists a homomorphism $\rho : I_K \to W$ and an I_K-equivariant splitting*

$$\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^2(X_k, \mathbb{Q}_\ell) \oplus V_\ell(-1) \tag{6.2}$$

where the I_K-action is trivial on the first summand, and on $V_\ell = \Phi \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$ it factors through $\mathrm{Gal}(L/K)$, where it is identified with the restriction of the reflection W-representation along ρ.

(ii) *L is the minimal Galois extension over which X_K has good reduction in the category of algebraic spaces. Moreover the isomorphism class of the I_K-representation $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$ depends, up to trivial summands, only on a formal Λ-isomorphism class of (X, x). In particular L depends only on this class.*

Proof. By Assumption $(*)$, Lemma 6.1 and Proposition 6.2 there exists a simultaneous resolution $\pi : \widetilde{X} \to X$ over some finite extension Λ'/Λ with $\pi_k : \widetilde{X}_k \to X_k$ the minimal resolution, and there exists a flat proper model X^{RDP}/Λ such that π factors through

$\alpha : \widetilde{X} \to X^{\mathrm{RDP}} \times \mathrm{Spec}(\Lambda')$, $X_K^{\mathrm{RDP}} \simeq X_K$ and X_k^{RDP} is a partial resolution of X_k, obtained by contracting the π_k-exceptional (-2) curves in $\widetilde{X}_k$; in particular X_k^{RDP} has finitely many rational double points and is otherwise regular.

Let $U_\ell = \Psi \otimes_{\mathbb{Z}} \mathbb{Q}_\ell$ and $T_\ell = V_\ell \cap U_\ell^\perp$. By the nearby cycles triangle (5.1), since X_k has a rational singularity and therefore the exceptional divisor in $\widetilde{X}_k$ is a tree of rational curves, we get

$$0 \to \mathrm{H}^2(X_k, \mathbb{Q}_\ell(1)) \xrightarrow{\mathrm{sp}} \mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell(1)) \xrightarrow{\phi} (\mathrm{R}^2\Psi_X \mathbb{Q}_\ell(1))_x \to \mathrm{H}^3(X_k, \mathbb{Q}_\ell(1))$$

Setting $E = \cup_i E_i \subset \widetilde{X}_k$ for the exceptional divisor, we have as in the proof of Theorem 5.16 that ϕ is identified with the restriction mapping

$$\mathrm{H}^2(\widetilde{X}_k, \mathbb{Q}_\ell(1)) \to \mathrm{H}^2(E, \mathbb{Q}_\ell(1)) \simeq \bigoplus_i \mathrm{H}^2(E_i, \mathbb{Q}_\ell(1)) \simeq V_\ell$$

and moreover it is surjective as the intersection matrix $(E_i \cdot E_j)_{ij}$ is nondegenerate ([Lip69, Lem. 14.1]). Consequently ϕ is surjective, $\mathrm{H}^i(X_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^i(X_k, \mathbb{Q}_\ell)$ for $i \neq 2$ has trivial monodromy action and, upon identifying sp with π_k^*, we have

$$\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell(1)) \simeq V_\ell^\perp \oplus V_\ell = \pi_k^* \mathrm{H}^2(X_k, \mathbb{Q}_\ell(1)) \oplus V_\ell$$

We can similarly apply the nearby cycles triangle (5.1) to X^{RDP}, noting that

$$\mathrm{R}^2\Psi_{X^{\mathrm{RDP}}} \mathbb{Q}_\ell \simeq \bigoplus_{x_i} (i_{x_i})_* (\mathrm{R}^2\Psi_{X^{\mathrm{RDP}}} \mathbb{Q}_\ell)_{x_i}$$

for $i_{x_i} : x_i \hookrightarrow X_k^{\mathrm{RDP}}$ the inclusions of the finitely many rational double points x_i. The α_k-exceptional curves of $\widetilde{X}_k \to X_k^{\mathrm{RDP}}$ are exactly the (-2)-curves E_i, whence the above argument yields

$$\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell(1)) \simeq U_\ell^\perp \oplus U_\ell = \alpha_k^* \mathrm{H}^2(X_k^{\mathrm{RDP}}, \mathbb{Q}_\ell(1)) \oplus U_\ell$$

Now the non-degenerate intersection pairings on Ψ and Φ yield $V_\ell = U_\ell \oplus T_\ell$ and $U_\ell^\perp = V_\ell^\perp \oplus T_\ell$ so $\mathrm{H}^2(X_k^{\mathrm{RDP}}, \mathbb{Q}_\ell(1)) \simeq T_\ell \oplus \mathrm{H}^2(X_k, \mathbb{Q}_\ell(1))$ equipped with the trivial I_K-action.

Applying Theorem 5.16 to X^{RDP}/Λ (in the form of Remark 5.17 (iii)) yields an I_K-equivariant splitting

$$\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell(1)) \simeq \mathrm{H}^2(X_k^{\mathrm{RDP}}, \mathbb{Q}_\ell(1)) \oplus U_\ell$$

where I_K acts trivially on the first summand and via restricting the reflection representation of W along a homomorphism $\rho : I_K \to W$ factoring through $\mathrm{Gal}(L/K)$, where L is the fixed field of $\ker(\rho)$. Thus we also have an I_K-equivariant splitting

$$\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell(1)) \simeq \mathrm{H}^2(X_k, \mathbb{Q}_\ell(1)) \oplus T_\ell \oplus U_\ell$$

with I_K acting trivially on T_ℓ and via ρ on U_ℓ. Twisting by (-1) yields (i), and (ii) proceeds exactly as Theorem 5.16 (ii). Note in particular if $\Psi = \emptyset$, meaning there are no (-2) curves in the minimal resolution, then the monodromy action is trivial and X already has good reduction over K. □

We note that the statements in Remark 5.17 also apply to this theorem, namely we can replace 'p is good for Ψ' with $p > 5$, and there is a variant of the theorem that holds when X_k has finitely many isolated rational singularities.

We also remark that Assumption $(*)$ is crucial in producing a simultaneous resolution of X/Λ after a finite extension. There are rational singularities and degenerations which deform outside the Artin component, and which don't admit simultaneous resolution after any finite base-change.

Example 6.4. Assume $p \neq 2$, let $\pi \in \Lambda$ be a uniformizer and set

$$Y = \{xy - z^2 = \pi w^2\} \subset \mathbb{P}^3_\Lambda$$

Let $\sigma : [x : y : z : w] \mapsto [x : y : z : -w]$ be the involution action and put $X \coloneqq Y/\langle\sigma\rangle$. Then X/Λ is a surface degeneration with $X_K \simeq \mathbb{P}^2_K$ and X_k has a singularity of type $\frac{1}{4}(1,1)$ at $x = [0 : 0 : 0 : 1]$. Indeed, after change of coordinates and completion one has $\widehat{\mathcal{O}}_{X_k,x} \simeq k[\![a,b]\!]^{\mu_4}$ where $\zeta \in \mu_4$ acts as $(a,b) \mapsto (\zeta a, \zeta b)$.

The minimal resolution of X_k is $\mathbb{F}_4 \to X_k$ where $\mathbb{F}_4$ is a Hirzebruch surface, locally the resolution of the cone over a rational quartic curve. Thus the singularity is rational. However, if X/Λ admitted a simultaneous resolution $\widetilde{X}$ after an extension Λ_L/Λ, we would have $\mathrm{H}^2(\widetilde{X}_{\overline{L}}, \mathbb{Q}_\ell) \simeq \mathrm{H}^2(\widetilde{X}_k, \mathbb{Q}_\ell)$ by smooth proper base-change. Since $\widetilde{X}_L \simeq \mathbb{P}^2$ and $\widetilde{X}_k \simeq \mathbb{F}_4$, this cannot happen as $h^2(\mathbb{P}^2, \mathbb{Q}_\ell) = 1 < h^2(\mathbb{F}_4, \mathbb{Q}_\ell) = 2$.

Note in this example that the ℓ-adic monodromy is trivial as $\mathrm{H}^2(\mathbb{P}^2_{\overline{K}}, \mathbb{Q}_\ell) \simeq \mathbb{Q}_\ell(-1)$ and I_K acts trivially on the Tate twist. What fails here specifically is the existence of a simultaneous resolution, as X/Λ deforms the singularity outside its Artin component. In general, we do not know whether every rational surface degeneration X/Λ necessarily yields finite ℓ-adic monodromy on $\mathrm{H}^2(X_{\overline{K}}, \mathbb{Q}_\ell)$.

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